The Incidence of Public Pay Reform:

Earnings, Sorting, and Service Delivery under Ghana's Single Spine Policy

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Abstract

Who gains when a government raises and compresses public pay, and does the raise reach the services the state delivers? Ghana's 2010 Single Spine Pay Policy placed roughly 400,000 public employees on a single salary schedule and raised the public wage bill by about half within two years. It was explicitly designed to lift the lowest-paid. It did not. Using four waves of nationally representative household surveys and a difference-in-differences identified within birth cohort, gender and ethnicity groups that no pay reform can reshuffle, I find that public earnings rose by about nine log points, but that the gain accrued almost entirely to workers with secondary education and above; those with middle schooling or less gained nothing measurable. The reform redistributed income within the public sector toward the educated rather than toward those it was meant to help. Nor did the money reach the services those workers staff. Matching household health surveys to districts' pre-reform public-employment shares, facility delivery, antenatal care and skilled birth attendance all rose by less where the state employed most, and the displaced births went home, unattended, rather than to private clinics. None of this is visible against the comparison group used in the prior literature, whose informal wages raced ahead during Ghana's commodity boom.

JEL codes: J31, J45, H83, I15, O12, J22.

Keywords: Public-private wage gap, incidence, pay compression, sorting, service delivery, difference-in-differences, Ghana, civil service reform.

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1 Introduction

Governments across the developing world routinely raise and compress public-sector pay, hoping to attract and retain the teachers, nurses and civil servants on whom public services depend. The instrument is almost always a salary spine: a single schedule that lifts the lowest grades, narrows the ratio of top to bottom pay, and is defended, in the legislation and in the press, as a policy for the low-paid. Whether it works that way is rarely tested.

Ghana's 2010 Single Spine Pay Policy is among the largest such reforms attempted. It placed roughly 400,000 public workers on one salary schedule and raised the public base-pay wage bill by about half within two years, in a country where that wage bill already absorbed close to half of tax revenue. A reform of this size is a substantial claim on public resources, and its defence rests on two propositions: that the money reaches the low-paid, and that better-paid workers deliver better public services.

Neither proposition follows from the design of a spine. A spine is a schedule, and what a schedule does to a particular worker depends on where she sits on it and on what the old arrangements had been paying her. By 2009 Ghana's eleven fragmented pay scales had eroded badly against the market, and they had eroded most at the grades requiring the most qualification, where outside options were strongest. Placement on the spine therefore closed a gap that was widest at the top of the ladder. A reform that raises the floor can still deliver most of its money well above it.

I study the reform using four waves of Ghana's national household survey spanning 1999 to 2017, and follow it beyond the payroll using four rounds of household health surveys. The comparison group is private employees on written contracts, who were excluded from the spine by statute but share the same economy, the same institutions and the same commodity cycle. Because a large pay increase changes who takes a public job, I identify the effect within groups defined by characteristics—birth cohort, gender and ethnicity—that no worker can acquire in response to a raise, so that the estimate comes from changes within a fixed group rather than from a changing one.

Public pay did rise. The public–private earnings gap is flat before the reform and about nine log points wider after it. But that finding depends entirely on which private workers form the comparison. Against all non-public workers—the conventional choice, and the one made by the closest prior study of this reform (Ampofo and Tchatoka, 2019, A&T)—the same reform appears to do nothing. The reason is visible in the raw data (Figure 2): between 1999 and 2017 mean nominal earnings rose about 28-fold for public workers and 76-fold for informal workers, so a comparison group containing

informal workers implies a counterfactual for public pay that no reform could beat.

The money did not reach the workers the spine was sold to help. The earnings gain rises with schooling: it is concentrated among workers with secondary education and above, and for those with middle schooling or less no treatment of the data delivers a gain that is detectable. Consistent with this, the public sector rewards a year of schooling about twice as richly as the private sector, and that advantage widened after the reform. A policy built to compress pay widened the distance between educated and uneducated public workers.

Nor did the money reach the services those workers staff. In the districts where the public sector was largest, facility delivery, antenatal care and skilled attendance at birth all rose by *less* than elsewhere, and the shortfall widened through 2022. Because the health surveys record where every birth took place, the mechanism can be identified rather than assumed: the births displaced from public facilities did not appear in private clinics but at home, and almost entirely without a skilled attendant.

The paper makes four contributions. First, it establishes who gains from a pay-compression reform. Spine systems that raise the floor and shrink top-to-bottom ratios are a standard instrument of civil-service policy, yet their incidence is rarely measured. The gains here accrue to the educated rather than to the low-paid the reform was sold to help, so a policy marketed as egalitarian can widen inequality among the workers it covers. This is a result about the *design* of public pay schedules rather than their average level.

Second, it follows a pay reform past the payroll to the services it was meant to improve. Such reforms are ultimately defended by what they do for the services the state delivers, yet the closest evidence here concerns the retention of health workers rather than the care households receive (Antwi and Phillips, 2013). Linking district exposure to household health data carries the question past the payroll to the citizen, and cautions that raising public salaries need not, on its own, buy better public services.

Third, it revisits the closest prior study of this reform. Ampofo and Tchatoka (2019) compare public workers to all non-public workers, across three survey rounds with a single post-reform observation, and conclude that the spine did not narrow the public–private gap. Both choices matter. Informal nominal wages raced ahead during the commodity boom, so a control pool containing them implies a counterfactual for public pay that no reform could beat; and one post-reform round cannot separate the immediate step in pay from what persisted. Measured against formal private workers, across four rounds, the same reform raised pay. Because much of the developing world's

workforce is informal and many of these economies run through commodity cycles, the lesson is not specific to Ghana.

Fourth, it offers an approach to evaluating reforms when the data are repeated cross-sections. Rather than adjusting for compositional change after the fact, I identify the effect from groups whose membership cannot respond to the policy, and show that the incidence result is stable across every treatment of composition while the pooled average is not. The approach applies wherever a policy changes who selects into the treated group and the data cannot follow individuals.

A simple framework organises these findings. Pay is assigned by grade; the gain from reform is the distance between the new schedule and what the old scales had been paying; and because a public post's grade is set by the qualification required to hold it, that distance widens with schooling. The reform's incidence should therefore follow qualification rather than need—a prediction the earnings, distributional and service-delivery results each test in turn.

This article is related to three literatures that have largely developed apart. The first estimates the public–private pay gap, with magnitudes that vary widely by country, period and comparison group (Gregory and Borland, 1999; Campos et al., 2017). The gap is small in rich economies (de Castro et al., 2013; García-Pérez and Jimeno, 2007) and more variable in developing settings, where public pay sits above *formal* private pay but below informal earnings for the unskilled (Panizza and Qiang, 2005; Jackman and Pauna, 1997; Adamchik and Bedi, 2000; Borjas, 2002); Kwenda and Ntuli (2018) and Younger and Osei-Assibey (2017) document sharp variation by skill in South Africa and Ghana. Two limitations run through this work: most estimates are cross-sectional, and the comparison group is treated as a modelling detail rather than an object of inference. The second is decisive, since a difference-in-differences is consistent only if the control would have followed the treated group's trend absent treatment (Abadie, 2005), and where informal wages grow quickly (Maloney, 2004; Saavedra and Chong, 1999) informal workers fail that test.

A second literature asks where in the distribution a pay intervention lands rather than how large it is on average. Institutional wage-setting compresses pay relative to the market (Freeman and Schettkat, 2001), and the quantile methods I use (Firpo et al., 2009; Powell, 2016) were developed to recover such effects; a related strand shows that the public sector narrows gender gaps (Anghel et al., 2011). What this work has not done is tie the incidence of a reform to the *structure* of the schedule that produces it.

A third literature links public pay to the quality of the state: higher offered wages

attract abler applicants (Dal Bó et al., 2013), pay interacts with intrinsic motivation in who selects into public jobs (Deserranno, 2019), and efficiency-wage models (Shapiro and Stiglitz, 1984; Akerlof, 1982; Akerlof and Yellen, 1990) motivate effort and retention responses, with Finan et al. (2017) surveying the field. Almost all of this evidence stops at the inputs to public service—recruitment, retention and effort. Whether higher salaries reach the services households actually receive is largely untested; the closest evidence for Ghana (Antwi and Phillips, 2013) concerns health-worker retention under an earlier adjustment, not delivery.

The paper proceeds as follows. Sections 2 and 3 describe the reform and the framework. Sections 4 and 5 present the data and the empirical strategy, Section 6 the results, and Section 7 the evidence on service delivery. Section 8 concludes.

2 Institutional Background

2.1 The problem the reform addressed

For the two decades before 2010, pay in Ghana's public service was set by some eleven separate salary structures, negotiated independently across ministries, agencies, and services. Workers doing comparable jobs in different services could therefore be paid on quite different terms, and the resulting inequities were a recurring source of grievance and industrial action (Baah and Reilly, 2009). Real public wages had also eroded through the structural-adjustment years, leaving many civil servants—teachers and health workers in particular—poorly paid relative to comparable formal-private workers by the mid-2000s (Debrah, 2010). The Single Spine Pay Policy (SSPP) was the government's response: a single, unified pay structure meant to deliver "equal pay for work of equal worth" across the entire public service (Government of Ghana, 2009).

2.2 What the reform did

The Fair Wages and Salaries Commission, created under the 2007 Fair Wages and Salaries Act, designed and administered the reform. A job-evaluation exercise placed every public-service job onto a single pay spine—one common ladder of twenty-five grades, or "levels"—in place of the eleven separate scales. Pay rises with the grade in fixed steps from a common base, so that a worker's salary is set by where her job sits on the one ladder rather than by which service bargained her scale. The SSPP took effect on 1 January 2010 and covered all workers paid under Article 190 of the 1992 Constitution—civil servants, teachers, health workers, the security services, and public

corporations, roughly 400,000 employees—who were migrated onto the spine in phases over the following two years.

Two features of the schedule shape the analysis that follows. First, the reform was redistributive in stated intent: it was designed to raise the pay of staff "benchmarked below median pay for the public service as a whole," while groups already paid above the median were expected to gain little (Cavalcanti, 2009). Second, the spine pays a raised floor and then rises by a roughly constant proportion—about two percent per pay point—with each step up the grade ladder, so pay climbs steeply across grades (the minimum annual basic salary was near GH¢ 1,100 at implementation). This grade-and-relativity structure is the one the framework in Section 3 builds on, and its incidence, it turns out, need not favour the low-paid once pay is measured in real terms.

The magnitude was first-order. Cavalcanti (2009) estimates that the SSPP raised the public base-pay wage bill from about GH¢ 1.9 billion in 2008 to GH¢ 2.8 billion by 2010—an almost 50 percent increase, and, in his words, a lower bound on the rise in total compensation. With the public wage bill already near 11.3 percent of GDP and 46 percent of tax revenue in 2008, the reform was a first-order fiscal commitment, not a marginal adjustment; the squeeze it placed on everything other than salaries is one lens on the service-delivery results of Section 7.

2.3 Why the reform provides usable variation

Three features make the SSPP useful for measurement. Its timing is plausibly exogenous to the private labour market: the policy was legislated in 2007 and implemented in January 2010, before Ghana's December 2010 commercial oil discovery, so it does not coincide with an oil-driven wage boom. Its coverage was statutory and nationwide, applying uniformly to public workers across regions and sectors. And formal private workers—those on written contracts—were excluded from the spine by construction, supplying a comparison group inside the same institutional and macroeconomic environment as treated workers.

Two institutional facts also supply the margins along which the reform can be measured. The first is that grade is assigned by job evaluation against the qualification a post requires, so a worker's schooling is the closest observable image of her position on the ladder; education groups are therefore not an arbitrary heterogeneity split but the empirical counterpart of the schedule itself. The second concerns geography. The spine raised a wage bill that already absorbed close to half of tax revenue, and it did so against public budgets in which salaries and the non-wage inputs of clinics and schools

compete for the same funds. The squeeze on those inputs is therefore mechanically larger where public employment is denser, which is what makes a district's pre-reform public-employment share a measure of exposure to the reform rather than merely a measure of how public it was. Section 7 builds that measure from the 2000 Population Census, a decade before the spine.

These features motivate the designs in Section 5, but they do not by themselves deliver clean identification: the GLSS is a repeated cross-section, so the composition of each sector can shift across waves, and the public–private comparison turns on who counts as a private-sector counterfactual. Sections 5 and 6.2 confront these concerns directly.

3 A Simple Framework for the Incidence of Spine Reform

The difficulty in evaluating a spine reform is not that the policy is hard to observe—it is statutory, dated and universal within the public sector—but that it operates on a variable no household survey records. A spine assigns pay by *grade*, and grade is what determines whether a given worker gains a little or a great deal. The framework below therefore does two things: it states what the schedule implies for incidence across grades, and it identifies the observable margin along which that implication can be tested.

The spine replaces eleven fragmented scales with a single schedule on which base pay rises by a roughly constant proportion with each step up the grade ladder:

$$\ln w_1(g) = \ln a + r\, g, \tag{1}$$

where g indexes the grade, a is the floor, and r the relativity between adjacent grades. Pay therefore climbs steeply with grade: a worker several steps up earns a large multiple of the floor, so the schedule is far from egalitarian even though it raises the minimum.

What a worker actually gains, however, is the difference between the new schedule and what the old scales had been paying her. Let $w_0(g)$ denote pre-reform pay at grade g. The gain from placement on the spine is

$$\Delta(g) \;=\; \ln w_1(g) - \ln w_0(g), \tag{2}$$

which I call the *erosion wedge*. Its shape is an institutional fact rather than an assumption.

The fragmented scales had lost ground against the market throughout the 2000s, and they had lost most at the grades requiring the most qualification, where outside options were strongest and retention hardest—the erosion that the reform was written to correct (Debrah, 2010; Government of Ghana, 2009). Hence $\Delta'(g) > 0$: the wedge is widest at the top of the ladder. A reform sold as lifting the floor has its incidence run the other way, because the workers who gain most are those high enough up the ladder that the old scales had failed them.

The wedge organises the empirical analysis. Deflated by a common price index, real gains grow with grade rather than shrink, so the reform should raise public earnings more for workers in higher-graded posts and should steepen the public sector's reward to qualification relative to the private sector's. Because such workers sit in the middle and upper-middle of the earnings distribution rather than at its floor, the gain should appear there rather than in the bottom decile. And because the wedge changes the level of pay at posts whose hours are set institutionally, it should raise pay by more than it raises hours. The framework is deliberately spare—equation (2) holds grade assignment fixed, which the empirical work then confronts directly—but it converts the question of incidence into a small set of signs to be checked against the data.

3.1 The spine as published

The framework is stylised, but its central premise—how steep the spine is and where workers sit on it—is observable, so the prediction can be grounded in the actual published schedule rather than left to the simple algebra above. Cavalcanti (2009) reports the statutory structure: twenty-five grades whose base pay follows $W(p) = 900 \times (1.02)^p$ in annual cedis, a constant two-percent relativity per pay point that runs from GH¢ 900 at the floor to about GH¢ 21,400 at the top—a nineteen-fold spread. This is exactly the log-linear schedule of equation (1); what converts it into a statement about incidence is where workers sit on it. Figure 1(a) plots the schedule.

What converts that schedule into a prediction about incidence is where workers actually sit on it. Cavalcanti's Figure 3 places the public workforce overwhelmingly in the upper-middle grades: Figure 1(b) reproduces the distribution, and it is far from uniform. About 71 percent of public workers fall in grades 8 through 15, the median worker sits at grade 11—already 1.5 log points, nearly five-fold, above the floor—and barely 3 percent occupy the three lowest grades the reform was marketed to lift. A spine that raises the floor therefore acts on a workforce that is almost entirely above it: the populated part of the ladder, and with it most of the reform's spending and most

of its proportional structure, lies in the middle, not in the thin low-paid tail. This is the calibrated counterpart of the signs the framework predicts, and Section 6.3 tests them against the GLSS.

Figure 1: The Single-Spine Schedule and Where Public Workers Sit on It



Notes: Panel (a) plots median annual base pay by single-spine grade (log scale); the schedule follows $W(p) = 900 \times (1.02)^p$, a constant two-percent relativity per pay point. Panel (b) plots the share of public workers in each grade (bars, left axis) and the cumulative share of the across-grade pay premium—log pay at a grade relative to the floor—that accrues at or below each grade (line, right axis); the dashed line marks the median worker's grade. Both the schedule and the employee distribution are taken from Cavalcanti (2009) (his Table 1 and Figure 3). The calibration is illustrative: the GLSS does not record a worker's spine grade.

3.2 From grades to observables

Equation (1) makes its prediction in grades, and no Ghanaian household survey records a worker's spine grade. Three mappings therefore carry the framework to what the data do record. Each rests on an institutional feature of the reform rather than on an assumption of convenience, and each turns an outcome that would otherwise look like a loose end into a derived prediction.

Grades and schooling. Placement on the spine was determined by job evaluation, and the evaluated grade of a public post is tied to the qualification required to hold it: a post requiring a degree enters the ladder several steps above one requiring a secondary certificate (Debrah, 2010; Cavalcanti, 2009). Write the assigned grade as $g = \gamma(s)$, increasing in schooling s. Substituting into equation (2), the erosion wedge facing a worker with schooling s is $\Delta(\gamma(s))$, with $\Delta'(\gamma(s))\,\gamma'(s) > 0$. Schooling is thus the observable image of grade, and the education gradient estimated below is not a heterogeneity

split chosen after the fact—it is the framework's central prediction, tested with the only proxy for grade the data contain. The same substitution delivers the prediction that the public sector's return to a year of schooling should steepen relative to the private sector's, since r applies to the grade ladder and γ converts schooling into grades.

Who holds a public job. The wedge also predicts a change in the workforce itself. If placement on the spine raised the return to qualification inside public employment, public jobs became more attractive to the educated, and the schooling composition of public employment should rise after the reform. This has an immediate implication for measurement: in a repeated cross-section, an estimator that holds schooling fixed removes part of the reform's own effect along with the confound, and should therefore return a *smaller* coefficient. The framework thus predicts not only the sign of the earnings effect but the direction in which composition-adjusted estimates will depart from it.

Gender. The framework contains no gender-specific mechanism, and this is a prediction rather than an omission. Women in Ghana's public sector are concentrated in nursing and teaching, occupations that occupy particular bands of the grade ladder. If the spine acts only through grade, any gender differential in the earnings gain must run through where women sit on that ladder, and should therefore attenuate once schooling is held fixed. A gender gap that survived conditioning on schooling would be evidence *against* the framework, not merely a result it fails to speak to.

Service delivery. The reform's downstream consequences follow from the government's budget rather than from the pay schedule. Let a public budget B divide between the wage bill W and the non-wage inputs N that make a clinic or school usable—drugs, fuel, equipment, maintenance—so that $B = W + N$, and let service output be $q = f(L, N)$ with both arguments productive. The spine raised W by roughly half within two years against a budget that did not expand correspondingly (Cavalcanti, 2009), so N must fall, and it must fall most where the public payroll absorbs the largest share of local public spending. With $\partial q/\partial N > 0$, service use should therefore rise *less* in districts with a larger pre-reform public workforce. This prediction motivates the district-level comparison reported below.

This mechanism also makes a sharper prediction than a general statement that services suffer. A budgetary squeeze on public provision pushes households out of public facilities, but says nothing that would draw them into private ones; a demand-side

story in which private provision expanded would. The two are distinguished by where displaced patients appear, which the health surveys record directly.

4 Data

The earnings and incidence results use the Ghana Living Standards Survey (GLSS). The service-delivery results combine the Demographic and Health Surveys with the 2000 Population Census, which together fix each district's pre-reform exposure to public employment; those data are introduced where they are used. All three sources are public, and the construction of every variable is documented in the data appendix.

4.1 The Ghana Living Standards Survey

The analysis uses four waves of the Ghana Living Standards Survey: Wave 4 (1998/99), Wave 5 (2005/06), Wave 6 (2012/13), and Wave 7 (2016/17). The GLSS is a nationally representative repeated cross-section of 8,700–15,000 households per wave drawn from all ten regions (Ghana Statistical Service, 2017), with detailed data on employment, earnings, hours, education, and household composition. Wave 5 is the pre-reform baseline, Waves 6 and 7 are post-reform, and Wave 4 provides a second pre-period for the parallel-trends placebo.

Earnings and hours. The outcome is log real monthly earnings, deflated to 2017 Ghana cedis with wave-specific CPI deflators from the Ghana Statistical Service (Wave 4 also requires the 2007 redenomination by 10,000). Earnings are winsorised at the 1st and 99th percentiles within wave. The secondary outcome is log weekly hours on the main job.

4.2 Sample construction

The treatment group is public-sector workers (central government, parastatals, the armed forces, and public education and health). The control group is formal private workers: those employed by a private firm under a written contract. Self-employed, agricultural, and informal wage workers are excluded from the main sample. The analytic sample is employed workers aged 15–64 with positive earnings; the four-wave sample contains 4,609 observations once the full control vector is required, and the pooled earnings sample used for the event study (which conditions on fewer controls) is larger.

4.3 Descriptive statistics

Table 1 reports pre-reform (Wave 5) means by sector for the 758 public and 493 formal private workers in that wave. Public workers are more educated (13.3 versus 11.9 years), more likely to be union members (72 versus 43 percent), and slightly older; in raw terms their log earnings are higher (6.78 versus 6.42). Because the two groups differ on observables, the specifications below condition on a full control vector and rely on the *change* in the gap rather than its level.

Table 1: Pre-Reform Descriptive Statistics (Wave 5, 2005/06)

	(1) Public		(2) Formal private	
	Mean	SD	Mean	SD
Log monthly earnings, 2017 GHc (winsorised 1-99 per wave)	6.780	0.828	6.422	0.901
Log weekly hours	3.740	0.442	3.852	0.438
Imputed years of schooling	13.299	2.575	11.853	2.543
Potential work experience (years)	22.177	10.948	18.729	11.366
1 = married or cohabiting	0.141	0.348	0.116	0.320
1 = trade union member	0.724	0.447	0.428	0.495
1 = female	0.297	0.457	0.254	0.435
Observations	758		493	

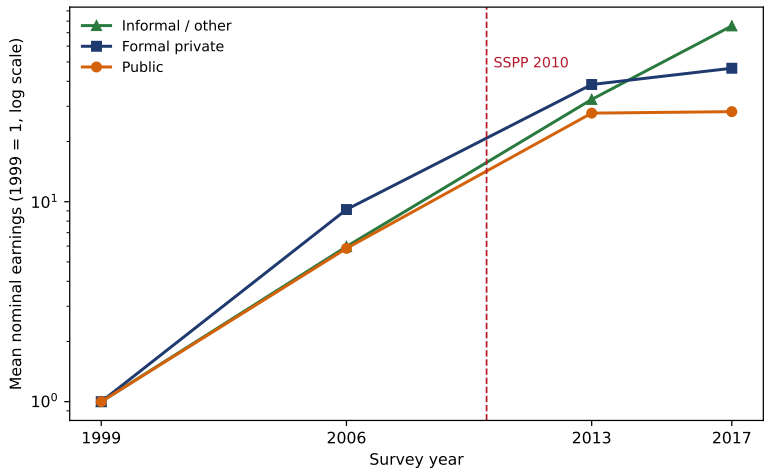
Notes: Wave 5 (2005/06) sample only. Formal private = written contract workers. Earnings in log 2017 GH¢ (real). Hours in log weekly hours. Education in years; experience in years; all others binary.

One raw fact frames the analysis. Figure 2 plots mean nominal monthly earnings by sector, indexed to 1999 on a log scale. Over 1999–2017 nominal earnings rose roughly 28-fold for public workers, 46-fold for formal private workers, and 76-fold for informal workers, whose pay nearly tripled in real terms as the commodity boom lifted the informal labour market from a very low base. Measured against formal private workers, who share the public sector's institutional environment, public pay pulls ahead after the reform; measured against a pool that is mostly informal, the same public pay appears *slow*. A reform can therefore appear ineffective solely because of the comparison against which it is measured.

Figure 10 plots log real earnings and log weekly hours by sector across the four waves. The series move together before 2010 and the earnings series diverge after it, while hours show no comparable break.

Because the GLSS is a repeated cross-section, the natural worry is that the composition of each sector shifts across waves. Table 22 reports observable means by sector and wave. Recomposition is modest but not absent: the public sector becomes somewhat more educated and more female over the period and its workers younger, while

Figure 2: Nominal Earnings by Sector, 1999–2017 (Indexed, Log Scale)



Notes: Mean nominal monthly earnings for public, formal private, and informal workers, each indexed to its 1999 value and plotted on a log scale, so that the slope of a line is its growth rate. Informal workers are employed earners who are neither public nor in formal private (written-contract) employment. Source: GLSS Waves 4–7. The dashed line marks the SSPP (January 2010).

the formal-private sample is broadly stable. The mild public-sector upskilling is worth flagging here because it is the same channel the selection bound in Section 6.2 addresses; the controls in the main specification absorb the observable part of it, and the Oster bound limits how much the unobserved part could matter.

Finally, the reform should show up in the public earnings distribution itself. Figure 11 plots the density of log real public earnings before the reform (2006) and after it (2013, 2017). The whole distribution shifts right after 2010, with the lower part moving most—the survey-data counterpart of a spine that raised the floor. This is the descriptive "first stage" behind the gap estimates: public pay did rise in real terms, and rose most where the spine lifted base pay by the largest proportion.

5 Empirical Strategy

5.1 The identification problem

The reform has a single statutory date, covers a group defined by law, and excludes formal private employees on written contracts. What it does not come with is a panel. The GLSS draws a fresh sample of households in each wave, so any comparison of public and formal private earnings before and after 2010 compares successive *samples* of workers rather than the same workers over time. If a large pay increase changes who takes and keeps a public job, part of the measured gap is a change in the workforce

rather than a change in pay.

In this setting that is not a hypothetical. Estimating the difference-in-differences with each worker characteristic in turn as the dependent variable, public employees gained 1.2 years of schooling and lost 2.3 years of potential experience relative to formal private employees after the reform, both precisely estimated; marital status and father's employment shift more modestly, and gender and household headship do not move (Appendix Table 23). The public workforce became more educated and younger. Because the public return to schooling exceeds the private return, a more educated public workforce raises measured public pay even if no individual's salary changed.

Higher public pay should draw better-qualified applicants into public employment, so this shift is precisely what the framework predicts. That is what makes it difficult to handle. The change in composition is part of the reform's effect and, at the same time, a source of bias in any comparison of successive samples: holding the public workforce at its pre-reform distribution of schooling would net out the response along with the confound. The design must therefore separate selection from treatment without conditioning away the channel through which the reform operated.

5.2 Preferred specification

Let $c(i)$ index a cell defined by worker i's birth-year band, gender and ethnicity, and let $s(i)$ index sector. I estimate

$$Y_{it} = \beta\,(\text{Public}_i \times \text{Post}_t) + \delta_t + \mu_r + X_{it}'\lambda + \theta_{c(i)s(i)} + \varepsilon_{it}, \tag{3}$$

where Y_{it} is log real monthly earnings, δ_t are survey-wave effects, μ_r region effects, X_{it} collects schooling, potential experience and its square, marital status, household headship and an indicator for the father holding formal employment, and θ_{cs} is a full set of cohort-by-sector fixed effects.

None of the three characteristics defining a cell can be chosen in response to a pay policy. The cell-by-sector effects absorb any time-invariant tendency of a given cohort to hold public rather than private employment, so β is identified from changes *within* a cohort-sector cell across waves: whatever made a particular type of worker more or less likely to hold a public job is differenced out, and what remains is the change in what that job pays. There are 72 cohort cells and 142 cohort-by-sector cells.[1] Dropping θ_{cs} and adding $\gamma\,\text{Public}_i$ recovers the conventional difference-in-differences, reported as a

[1] Cells with a single observation are dropped, which costs 52 of the 8,206 observations in the estimation sample.

benchmark in column 3 of Table 2.

The design also fixes the level of inference. The identifying variation sits at the cohort-by-sector level, so cohort is the natural clustering dimension, and its 72 clusters avoid the small-cluster problem that ten regions create (Cameron and Miller, 2015). I report region-clustered and wild cluster bootstrap standard errors alongside.

5.3 Threats

Three threats bear on equation (3). *Differential pre-trends* would show up in the 1999 placebo, which the event study tests directly and cannot reject. *A contemporaneous shock*—chiefly Ghana's late-2010 oil discovery—is absorbed by wave effects to the extent that it raised all formal wages symmetrically; Section 6.2 takes up the possibility that the commodity boom moved the control group's pay for reasons of its own, and shows that removing the boom regions raises rather than lowers the estimate. *Time-varying selection within a cell* is what the design cannot remove. No estimator applied to repeated cross-sections can, which is why Section 8 identifies individual payroll records as the decisive next step.

Two features of the setting keep the design on firm ground. The reform has a single implementation date and two groups, so the negative-weighting problems that arise under staggered adoption (Goodman-Bacon, 2021; Roth et al., 2023) do not apply. And the conclusion does not rest on the pooled average alone: the cohort pseudo-panel in column 6 of Table 2 and the incidence result are estimated on different aggregations of the same data and point the same way (Abadie, 2005; Roth et al., 2023).

6 Results

6.1 Main Effects

Table 2 reports the effect of the reform on public pay and, in the same columns, shows how much of that effect survives the identification problem of Section 5.1.

Table 2 reports the effect of the reform on the public–private earnings gap, moving from the naive comparison to the preferred design on a single fixed sample of 8,206 workers, so that what changes across columns is the specification rather than the sample.

The raw difference-in-differences puts the gap at 21 log points (column 1), and survey-wave and region fixed effects leave it there (column 2). The control vector brings it to 12 log points (column 3). That is the conventional specification, and the one the

composition evidence calls into question. Adding cohort fixed effects leaves 13 log points (column 4). The preferred specification, which interacts those cells with sector and so absorbs any time-invariant tendency of a cohort to hold public rather than private employment, gives 8.9 log points (column 5)—a gain of about nine percent on a pre-reform mean of 6.78 log points, and the part of the conventional estimate that does not rest on who holds a public job.

The estimate sits near the conventional threshold. Clustered by cohort, the p-value is 0.040 and the wild cluster bootstrap gives 0.049, with a 95 percent confidence interval from 0.001 to 0.176; clustering on regions widens these to 0.054 and 0.084.

Column 6 collapses the same cells to means and tracks them across waves, so that the unit of observation is a fixed group rather than a changing set of individuals. That estimate is 16.2 log points and is significant at the five percent level. The two designs share no sample construction and bracket the conventional estimate from either side.

Panel B repeats the ladder with log weekly hours as the outcome, and here the framework's prediction is not borne out. Under the benchmark specification hours are flat, as earlier drafts of this literature would lead one to expect. Under the preferred specification they rise by 6.3 log points, significant at the ten percent level. The point estimate is the same in every column; what changes is precision, and the gain comes from clustering at the level of the identifying variation rather than on ten regions.

This changes the interpretation of the earnings effect. Estimated on the sample for which both outcomes are observed, monthly earnings rise by 9.6 log points and hours by 6.3, so the implied effect on the hourly rate of pay is 3.3 log points. Roughly two-thirds of the measured gain in monthly earnings is more time worked rather than a higher rate of pay. Monthly earnings remain the right outcome for a reform that sets monthly salaries, and it is that quantity the spine schedule governs; but the reform did not raise pay at unchanged effort.

6.1.1 Dynamics

A single coefficient cannot show whether the two sectors were already diverging. The event study can, and it is the most direct evidence that the reform, rather than a pre-existing trend, produced the gap.

Figure 12 presents two event studies that use the actual GLSS interview year on the horizontal axis and 2006 as the omitted reference. Panel A exploits cross-regional exposure: regions with a pre-reform (Wave 5) public employment share above the median (≥ 0.383) are coded high-exposure, and the specification interacts exposure with

Table 2: The Effect of the Single Spine on Public–Private Earnings and Hours

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Log real monthly earnings						
Public × Post	0.213***	0.211***	0.115***	0.134***	0.089**	0.162**
	(0.047)	(0.042)	(0.036)	(0.034)	(0.043)	(0.065)
Observations	8,206	8,206	8,206	8,158	8,154	88
Panel B. Log weekly hours						
Public × Post	0.056	0.058	0.058	0.049	0.063*	—
	(0.050)	(0.046)	(0.045)	(0.048)	(0.033)	
Observations	7,708	7,708	7,708	7,665	7,660	
Wave and region FE	No	Yes	Yes	Yes	Yes	Wave
Controls	No	No	Yes	Yes	Yes	Yes
Cohort fixed effects	No	No	No	Cohort	Cohort × sector	Cohort × sector
Clustering	Region	Region	Region	Region	Cohort	Cohort
Unit of observation	Worker	Worker	Worker	Worker	Worker	Cohort cell

Notes: Columns 1–5 are estimated at worker level on a fixed sample, so that differences across columns reflect the specification rather than the sample; Panel B rests on fewer observations because weekly hours are unrecorded for some workers with valid earnings. Column 1 is the raw difference-in-differences; column 2 adds survey-wave and region fixed effects; column 3 adds the control vector and is the conventional specification, reported as a benchmark; column 4 adds fixed effects for cohort cells defined by birth band, gender and ethnicity; column 5, the preferred specification, interacts those cells with sector, absorbing any time-invariant tendency of a cohort to hold public rather than private employment. Columns 4 and 5 lose a small number of observations to singleton cells. Column 6 collapses the same cells to means and treats the cell as the unit of observation across waves (Deaton, 1985), weighting by cell size and dropping cells with fewer than twenty workers; it is not estimated for hours. Standard errors are clustered at the level shown, which is the level of the identifying variation in each column.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

survey-year dummies on all workers with positive earnings ($N \approx$ 36,600). Panel B is the individual-level DiD, interacting Public_i with survey-year dummies on the public and formal private sample ($N \approx$ 9,068).

In the difference-in-differences (Panel B), the public–private gap is flat and statistically indistinguishable from zero in every year before the reform, then opens once the spine takes effect (coefficients in Table 8, appendix). The regional-exposure design in Panel A tells the same story from independent variation: regions with a larger pre-reform public workforce pull away from less exposed regions only after 2010. The pre-reform coefficients are the parallel-trends test, and none is significant in the difference-in-differences; in the exposure design the 2005 coefficient is marginally significant at ten percent, a one-year blip against the 2006 reference rather than a trend. The hours margin shows no comparable dynamics: its event study (Figure 13) is flat in every year under both designs, the dynamic counterpart of the null hours effect reported above.

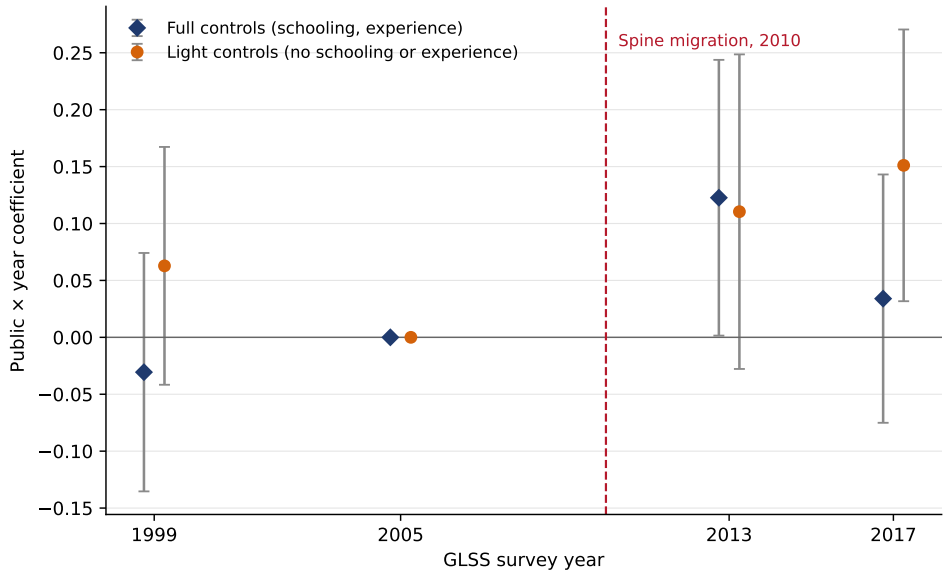
The event studies above use the baseline design. Because the pre-trend evidence and the headline estimate should come from the same specification, Figure 3 repeats

the exercise under equation (3), with GLSS5 (2005) as the omitted reference.

Two features are notable. First, the pre-trend is clean: the 1999 coefficient is -0.031 with a standard error of 0.053, and a formal test that it is zero returns $p = 0.57$. Under the preferred design there is no sign that the two groups were diverging before the reform. Second, the two series in the figure—one conditioning on schooling and experience, one conditioning only on pre-determined demographics—lie on top of each other before the reform and separate after it. Conditioning only on demographics, the gap reaches 15 log points by 2017; adding schooling and experience leaves 3. The divergence is informative rather than anomalous: the part of the gap that opened by 2017 is carried by the education composition of the public workforce. The same fact appears in Table 30 as the reweighting result and again in the education gradient. Three independent routes deliver the same conclusion.

I also report what the design does not deliver. Estimated with GLSS5 as the sole pre-period, the pooled coefficient is 0.070 with a standard error of 0.055: the magnitude is stable, within one standard error of the four-wave estimate, but a single pre-period does not identify it precisely. The effect is not fading, either—estimated against 2013 alone the coefficient is 0.088, and against 2017 alone 0.074.

Figure 3: Event Study under the Preferred Design



Notes: Coefficients on Public$_i$ interacted with survey year from equation (3), with cohort-by-sector, wave and region fixed effects; 2005 is the omitted reference. The upper series conditions on pre-determined demographics only (gender, marital status, household headship, father in formal employment); the lower series adds years of schooling and potential experience. Bars are 95 percent confidence intervals from standard errors clustered by cohort. The dashed line marks the start of migration onto the spine in January 2010.

6.1.2 Persistence

Tracing the earnings effect across horizons under the formal-private control (Table 24) shows that it is significant and persistent. Against the pooled 1999–2005 baseline, the public–private premium is about 0.18 by 2013, 0.12 over the full window, and 0.11 in a long difference that compares the pre-reform waves with 2017 alone—all individually significant. The effect appears within a few years of implementation and does not fade over the following decade; if anything the early-horizon estimate is the largest.

6.1.3 Labour supply

Panel B of Table 2 showed hours rising by 6.3 log points under the preferred design. Because the reform also changed wages sharply, it can be used to ask whether that is a labour-supply response, and Table 29 reports the attempt. Using the reform as an instrument for wages, the first stage is adequately strong—a Kleibergen–Paap F of about 11.5, above the conventional threshold—and the implied Frisch elasticity is about 0.5 with a standard error of roughly 0.4.

That estimate is imprecise, and on its own it cannot reject zero. But it is consistent in sign and in magnitude with the reduced form: an elasticity of 0.5 applied to a nine-log-point earnings increase implies four to five log points of additional hours, against the 6.3 observed. The instrumented and direct evidence therefore point the same way, toward a modest positive intensive-margin response rather than the null that jobs with institutionally set schedules would lead one to expect. Two readings are available and the data do not separate them: public workers may genuinely have supplied more effort once the spine raised the return to holding the post, or reported hours may have risen with the formality of the employment relationship the reform documented. Either way, the earnings gain is not a pure change in the price of a fixed quantity of work.

6.2 Robustness

An effect that appears only against one comparison group, or only under one treatment of composition, is not an effect. The most consequential of those choices is the comparison group, and it is where the prior literature reached a different conclusion from this one.

6.2.1 The control group

The choice of comparison group is the single most consequential decision in this setting, so I treat it as a result rather than a preliminary. Ampofo and Tchatoka (2019) compare public workers to *all* non-public employed workers—a group that pools informal wage earners, the self-employed, and agricultural workers with formal private employees. Informal nominal earnings in Ghana grew rapidly over the 2000s commodity boom from a low base, so an informal-inclusive control implies a fast counterfactual growth path for public pay and mechanically shrinks or reverses the estimated effect.

Table 25 holds the formal-private control fixed and varies the estimation window and the earnings measure. Every window that uses all four waves returns a significant estimate; the shorter post-2006 window is positive but imprecise. The result is not an artefact of deflation, since nominal earnings—the measure A&T adopt—give a similar premium against the same control.

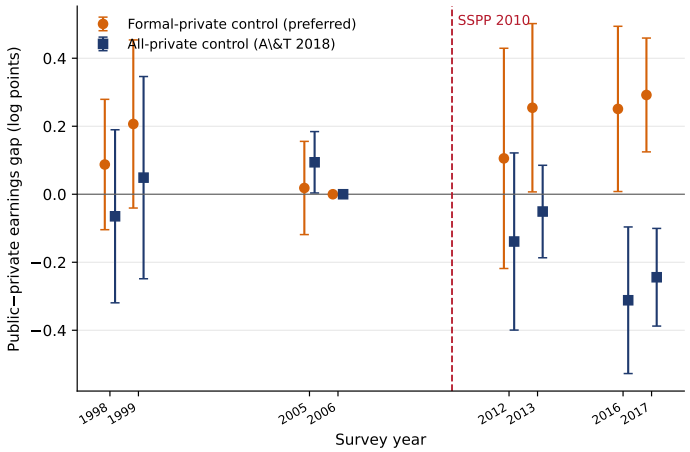
Figure 4 displays the comparison directly. Against formal private workers the gap is flat before the reform and opens after it; against all non-public workers—the A&T choice—the same reform appears to *lower* relative public pay after 2010. The two series share the treatment, the outcome and the specification, and differ only in who counts as the comparison worker. Appendix A reports what their design yields in these data: an effect indistinguishable from zero.

The lesson generalises beyond Ghana. Where the informal sector is large and its nominal wages grow quickly, the formal public-sector pay gap cannot be measured against an informal-inclusive control, because the control group is itself on a steep and unrelated trend; the comparison must be drawn within the formal labour market that the treated workers inhabit.

Formal private workers are a better counterfactual than all private workers, but they are not immune to shocks of their own: commercial oil production began in December 2010 and gold mining ran throughout. Table 31 applies the obvious corrections—absorbing industry-by-wave and region-by-wave effects, restricting both sectors to services, and dropping the extractive industries—and the estimate survives all of them, at between 6 and 8 log points, less precisely because the industry variable is coarse and missing for part of the sample.

The more revealing exercise goes in the opposite direction. Excluding Western region, where the offshore oil industry came ashore, raises the estimate to 12.5 log points, and excluding Ashanti's gold-mining economy as well raises it to 14.9. The explanation lies in the comparison group itself: over the reform period formal private log earnings

Figure 4: The Same Reform, Two Control Groups



Notes: Each series plots the coefficients from a difference-in-differences event study of log real monthly earnings on the public-sector indicator interacted with survey year, with 2006 omitted; bars are 95% confidence intervals. The two series use the identical outcome, treatment, controls (female, age, marital status, household headship, father in formal employment), and region and survey-year fixed effects, and differ only in the comparison group: formal private workers (the preferred control) versus all non-public workers (the control used by Ampofo and Tchatoka, 2019). Standard errors are clustered by region. The dashed line marks the SSPP (January 2010).

grew by 1.34 in trade and services and by 0.73 in agriculture, but by -0.03 in the service activities where public employment is concentrated. The boom inflated parts of the comparison group, which biases the estimate *down*. Contamination of the control group is therefore a reason the headline understates the reform, not a reason to doubt it.

6.2.2 Composition and alternative estimators

Composition can also be treated as a nuisance to be corrected rather than designed around, and four such corrections are available. Reweighting post-reform observations to the pre-reform covariate distribution (DiNardo et al., 1996) leaves the estimate at 7.9 log points when pre-determined demographics are balanced and 4.8 when experience is added; the weights are well behaved, retaining between 62 and 71 percent of the nominal sample. Balancing schooling as well drives the coefficient to 2.6, but that specification removes the reform's own response along with the confound and is a lower bound rather than a cleaner estimate. The doubly-robust estimator of Sant'Anna and Zhao (2020) returns 0.20; the bound of Oster (2019) implies that selection on unobservables would have to be nearly twice as strong as selection on observables to eliminate the effect; and the relative-magnitudes analysis of Rambachan and Roth (2023) shows the long-run estimate surviving violations of parallel trends up to three-quarters of a

pre-reform deviation that is itself small and insignificant. With one usable pre-period the last of these is a weak restriction, reported for transparency rather than as a claim. All four are collected in Appendix D, and all carry less weight than the designs of Section 6.1, which do not require composition to be modelled correctly at all.

Taken together, the corrections move the estimate in opposite directions and none reverses its sign. Those that address contamination of the comparison group raise it, to between 12 and 15 log points; those that adjust for composition lower it, to between 5 and 8; the alternative estimators sit at 16 to 20. The preferred estimate of 8.9 log points lies near the middle of that range and below most of it.

Two further checks concern the regional-exposure design and the event study. The exposure result is not an artefact of dichotomising exposure: a continuous pre-reform public-employment share and a leave-one-out version of it, which strips any mechanical link between a worker's own sector and her region's exposure, both reproduce the growing post-reform pattern and flat pre-trends (Table 16). Greater Accra is itself a *low*-exposure region, so the design is not tracking a capital-city trend. Re-estimating the event study under minimal controls, an added age-squared term, and the exclusion of Wave 4 leaves the post-reform path intact; it reverses only under the informal-inclusive control group (Figure 17), which is the control-group result arriving a second time.

6.3 Incidence

The average conceals the question the reform was sold on. A spine that lifts the floor is defended as a policy for the low-paid; the framework of Section 3 predicts the opposite, because the erosion wedge is widest at the top of the ladder and the gain should therefore rise with schooling. This is the paper's central test, and the evidence for it is more robust than the average it decomposes.

6.3.1 The education gradient

The average premium masks the workers it reaches, and the incidence is the more robust finding. Table 3 estimates the difference-in-differences separately within education groups, under each of the three treatments of composition used above.[2] Composition moves far less inside an education group than across the whole sample, so these estimates are much less exposed to the problem of Section 5.1 than the pooled average

[2]The education groups are built from imputed years of schooling rather than the survey's education-level code. That code does not distinguish the two highest categories in GLSS7—the fourth level is empty in that wave—so a stratified analysis based on it would compare different populations across waves. Merging the two highest levels restores comparability.

is.

The gain rises with schooling and is concentrated above the middle. For workers with secondary schooling the effect is between 16 and 19 log points and is similar under all three treatments; for the tertiary-educated it is between 21 and 28. For workers with middle schooling or less, no treatment delivers an effect that is stable in sign, so the bottom of the distribution shows no detectable gain rather than a loss. A test that the four coefficients are equal rejects under the baseline ($p = 0.014$) and under the preferred design ($p = 0.035$). The spine was built to lift the floor; the floor is where the least evidence of a gain is.

Figure 5 shows the same result dynamically. The secondary group is the cleanest: flat before the reform, positive after it. The tertiary group carries a qualification worth stating plainly—its 1999 coefficient is -0.211 with a p-value of 0.064, so tertiary public workers were already below trend before the reform and part of their post-reform gain is recovery from that base. The secondary-group result does not depend on this and is the one on which the incidence claim principally rests.

Table 3: The Incidence of the Reform Across the Education Distribution

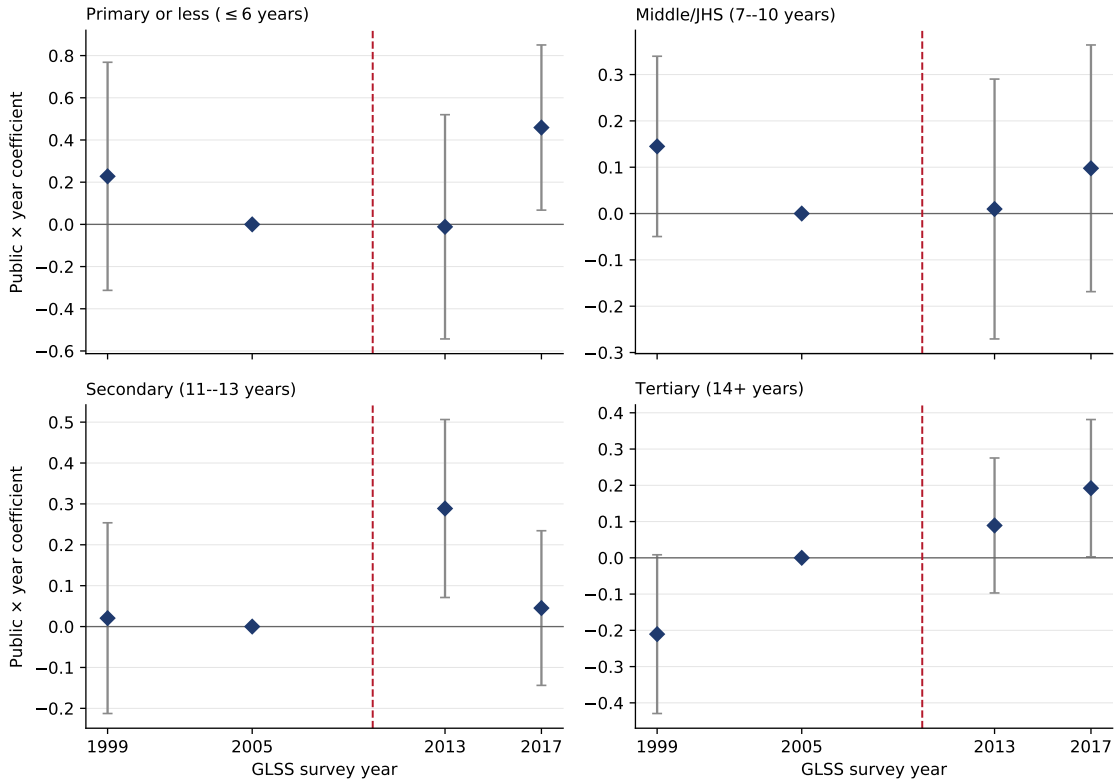
	Baseline controls (1)	Composition-reweighted (2)	Cohort × sector fixed effects (3)	Workers (4)
Primary or less (≤6 years)	-0.283	-0.253	0.220	739
	(0.183)	(0.156)	(0.237)	
Middle/JHS (7–10 years)	-0.116**	-0.167**	-0.021	2,673
	(0.051)	(0.052)	(0.090)	
Secondary (11–13 years)	0.188**	0.166*	0.164*	1,666
	(0.073)	(0.077)	(0.075)	
Tertiary (14+ years)	0.282***	0.213**	0.213**	3,128
	(0.083)	(0.081)	(0.088)	
Joint test of equal effects	$F = 6.25$, $p = 0.014$ (col. 1); $F = 4.46$, $p = 0.035$ (col. 3)			

Notes: Each cell reports the Public × Post coefficient from a difference-in-differences estimated separately within an education group. Education groups are built from imputed years of schooling rather than the survey's education-level code, because that code does not distinguish the two highest categories in GLSS7 and is therefore not comparable across waves. Column 1 uses the baseline control vector; column 2 reweights post-reform observations within sector to the pre-reform distribution of demographics and experience; column 3 adds cohort × sector fixed effects. Composition moves far less within an education group than across the whole sample, and the gradient is stable across all three treatments. The joint test asks whether the four coefficients are equal. Standard errors clustered by region in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Splitting instead by sex (Table eftab:skill), the gain is 0.247 log points for women and 0.075 for men, the latter indistinguishable from zero. The price of education moves the same way (Table eftab:edureturns): the public sector rewards a year of schooling

Figure 5: Event Study by Education Group, Preferred Design



Notes: Equation (3) estimated separately within each education group, with 2005 the omitted reference. Bars are 95 percent confidence intervals from standard errors clustered by cohort. The dashed line marks the start of migration onto the spine in January 2010.

about twice as richly as the formal private sector, and the public-sector earnings premium roughly doubled across the reform. A triple-difference interacting the treatment with years of schooling (Table 28) confirms that the public sector's education gradient steepened after the spine took effect. The reform did not merely raise public pay; it tilted the public pay schedule toward the educated.

6.3.2 Distributional effects

The framework predicts that the gain should appear in the middle of the distribution, where educated public workers sit, rather than at its floor. Quantile difference-in-differences estimates bear this out (Table 9): the public–private gap widens in the lower-middle of the distribution, significantly at the 25th and 50th percentiles before controls and at the 25th with the full control vector, but not at the 10th percentile or at the top. The companion within-sample quantile regression (Table 10) tells the same story from the other side, with a significant gain at the median, none at the top, and if

anything a negative coefficient at the bottom decile. This is what the framework implies once the workforce is mapped onto the spine: the ladder raises pay proportionally more at higher grades, but public workers cluster in the middle and upper-middle grades and thin out at the very top (Figure 1), so realised gains concentrate in the middle rather than at the ceiling.

6.3.3 Sorting

A repeated cross-section cannot follow workers across sectors, so net reallocation of skilled workers is not directly identified. What the data support is a statement about prices: the reform raised the relative reward to education inside the public sector (Table 27 and the triple-difference above), the signal that would over time make public employment more attractive to educated workers. The composition evidence of Section 5.1 is the quantity counterpart—public employment did become more educated—but because that shift is also the confound the design must neutralise, it is not a clean estimate of sorting. Separating the two requires data that track individuals.

6.3.4 Gender and sub-sector event studies

Re-estimating the event study by sex (Figure 15) shows larger post-reform gains for women, though the female series carries a significant 1999 pre-reform coefficient, so those estimates are not cleanly identified; the male series has flat pre-trends and a smaller effect. Splitting public workers into education-and-health and public-administration groups (Figure 16) shows a clearer post-reform path for teachers and health staff, who were among the lowest-paid civil servants before the spine and whose grades it raised most.

7 Did the Reform Reach Service Delivery?

The incidence results describe who, among public workers, the spine reached. A pay reform of this size is ultimately justified by what it does for the public *services* citizens receive: better-paid and better-retained nurses, midwives, and teachers are supposed to translate higher salaries into care that shows up in households. Whether the spine reached that far is an open question. The closest prior evidence studies the retention of health workers under an earlier salary adjustment rather than the services delivered under the spine itself (Antwi and Phillips, 2013). I bring four rounds of the Demographic

and Health Surveys—2003 and 2008 before the reform, 2014 and 2022 after it—to bear on it.

The design mirrors the regional-exposure logic of the earnings analysis at a far finer grain. For each of about a hundred districts I measure the pre-reform intensity of public employment from the 2000 Population Census—the share of working residents employed in the public sector—and treat it as the district's exposure to a reform that raised public pay. Each DHS cluster is placed in its district from its GPS coordinates, so that a mother's exposure is fixed by where she lives and by the local public workforce a decade before the spine. I then ask whether service use rose more, after 2010, in districts where the public sector loomed larger, controlling for district and survey-year fixed effects. The outcomes are whether a birth took place in a public health facility, whether the mother had at least four antenatal visits, and whether a skilled provider attended the birth.

The answer is not the optimistic one. Public services expanded everywhere over these two decades—facility delivery rose from two-fifths to nearly three-quarters of births—but that expansion was *weaker*, not stronger, where the public sector was largest. Districts more exposed to the reform saw public-facility delivery, antenatal care and skilled birth attendance all rise by less than low-exposure districts, and the shortfall widened through 2022 (Table 4 and Figure 6); the pre-reform path is flat, so this is a divergence that opens after the spine rather than a trend that predates it. All three effects are significant at the one percent level, and they point the same way.

Two features discipline what can be inferred. The effect is an urban phenomenon—present in towns and cities, absent in rural clusters—and it survives the controls that would absorb the most obvious confounds: a full set of urban-status-by-year interactions and, more demandingly, region-by-year fixed effects that identify the coefficient only from differences across districts within the same region and year (Table 19, in the appendix).

The mechanism can be identified rather than assumed. Two hypotheses are consistent with a fall in public-facility delivery: *substitution*, in which a growing private sector drew deliveries away with no loss of care, and *crowd-out*, in which the squeeze on non-wage inputs predicted by Section 3.2 made public clinics harder to use. They differ in where the displaced births went—to private facilities under the first, home under the second.

The DHS records place of delivery exhaustively, so the predictions can be tested directly. Table 5 re-estimates the same difference-in-differences with each destination

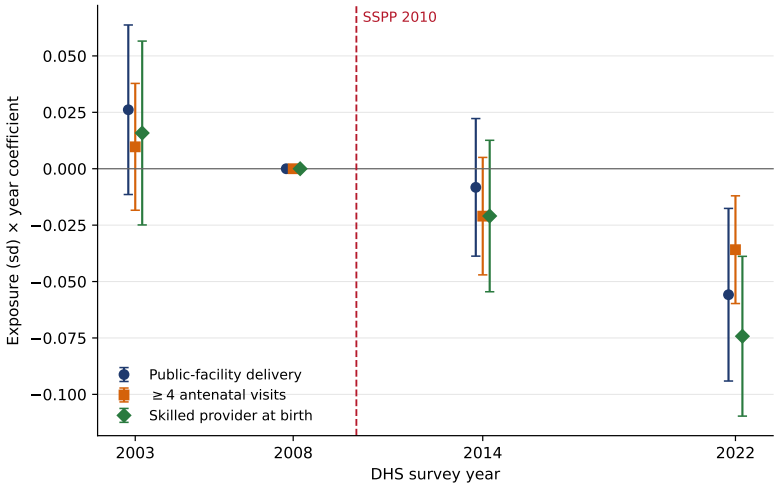
Table 4: Public-Employment Exposure and Public Service Delivery

	Public-facility delivery	ANC (≥4) visits	Skilled birth attendant
Exposure (sd) × Post	-0.047***	-0.034***	-0.057***
	(0.016)	(0.011)	(0.017)
Pre-reform mean (2003–08)	0.413	0.744	0.481
Post-reform mean (2014–22)	0.724	0.872	0.784
Observations	14,198	14,089	14,297
District FE	Yes	Yes	Yes
Survey-year FE	Yes	Yes	Yes
Districts (clusters)	102	102	102

Notes: Difference-in-differences across four Ghana DHS rounds (2003 and 2008 pre-reform; 2014 and 2022 post-reform). The sample is the most recent birth of each woman. *Exposure* is the standardised pre-reform (2000 Population Census) public-employment share of a mother's district of residence, assigned from cluster GPS coordinates; *Post* equals one in 2014 and 2022. Public-facility delivery is birth in a government/public health facility; ANC is at least four antenatal-care visits; skilled birth attendant is delivery attended by a doctor, nurse or midwife. All columns include district and survey-year fixed effects and are weighted by DHS sampling weights.

Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6: District Public-Employment Exposure and Service Delivery



Notes: Each series plots the exposure (sd) × survey-year coefficient from a difference-in-differences across four Ghana DHS rounds, with 2008 (the last pre-reform round) as the omitted reference and 95% confidence intervals. Exposure is the standardised pre-reform (2000 Census) public-employment share of the mother's district, assigned from cluster GPS coordinates; district and survey-year fixed effects are absorbed and standard errors are clustered by district (102 districts). The dashed line marks the 2010 implementation of the Single Spine Pay Policy.

share as the outcome, on a common sample of births with a valid response, so that the coefficients sum to zero by construction and the decomposition is exact. The private-facility share does not move: its coefficient is -0.005 with a standard error of 0.008. The home share rises by 5.1 percentage points, significant at the one percent level and almost exactly offsetting the fall in public-facility delivery. Substitution toward private providers is not what happened. Births displaced from public facilities were delivered at home.

Two further results confirm this. The rise is almost entirely in *unattended* home births—the home-and-unskilled share rises by 5.2 percentage points while the home-and-skilled share does not move—which is why skilled attendance falls alongside facility delivery. And the pre-reform coefficients on all four shares are indistinguishable from zero, so the reallocation begins after the spine.

Table 5: Where Did the Births Go? Decomposing Place of Delivery

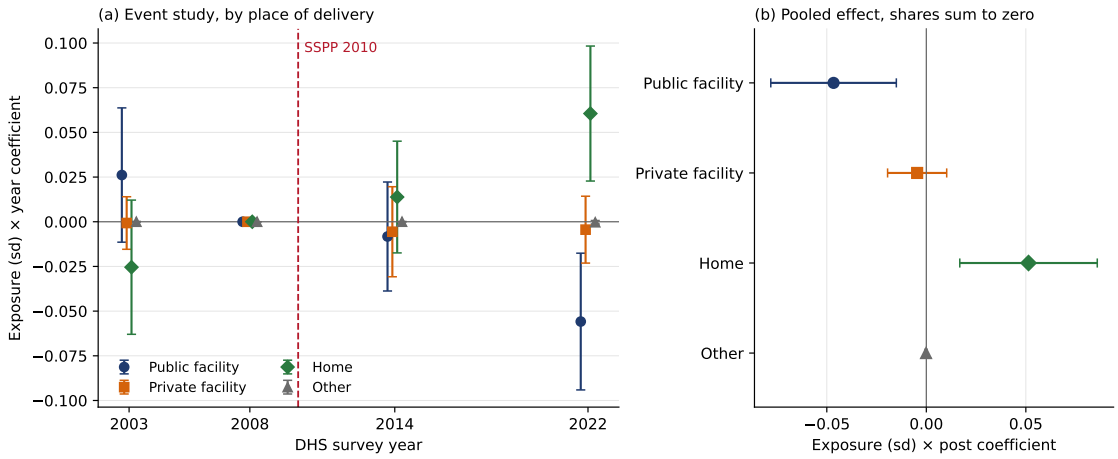
	Public facility (1)	Private facility (2)	Home birth (3)	Other (4)
Exposure (sd) × Post	-0.047***	-0.005	0.051***	-0.000
	(0.016)	(0.008)	(0.018)	(0.000)
+ urban × year	-0.033**	-0.011	0.044***	-0.000
	(0.015)	(0.008)	(0.016)	(0.000)
Region × year FE	-0.037**	0.013*	0.024*	0.000
	(0.018)	(0.007)	(0.014)	(0.000)
Pre-reform mean (2003–08)	0.413	0.083	0.505	0.000
Post-reform mean (2014–22)	0.724	0.071	0.203	0.002
Observations	14,198	14,198	14,198	14,198
District FE	Yes	Yes	Yes	Yes
Survey-year FE	Yes	Yes	Yes	Yes
Districts (clusters)	102	102	102	102

Notes: Each column reports a separate difference-in-differences for one destination share, estimated on a common sample of births with a valid place-of-delivery response, so the four coefficients sum to zero by construction. The specification is identical to Table 4: *Exposure* is the standardised pre-reform (2000 Population Census) public-employment share of the mother's district, assigned from cluster GPS coordinates, and *Post* equals one in the 2014 and 2022 rounds. Rows two and three report the same coefficient adding urban-status × survey-year interactions and replacing survey-year with region × survey-year effects. All specifications include district fixed effects and DHS sampling weights. The decomposition distinguishes crowding out, under which births displaced from public facilities appear at home, from substitution toward private providers, under which they appear in private facilities.

Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The downstream evidence points the same direction as the incidence results. A spine built to raise the floor delivered its earnings gains to educated public workers above the middle of the distribution; it did not, over the following decade, pull citizens toward

Figure 7: Where Did the Births Go?



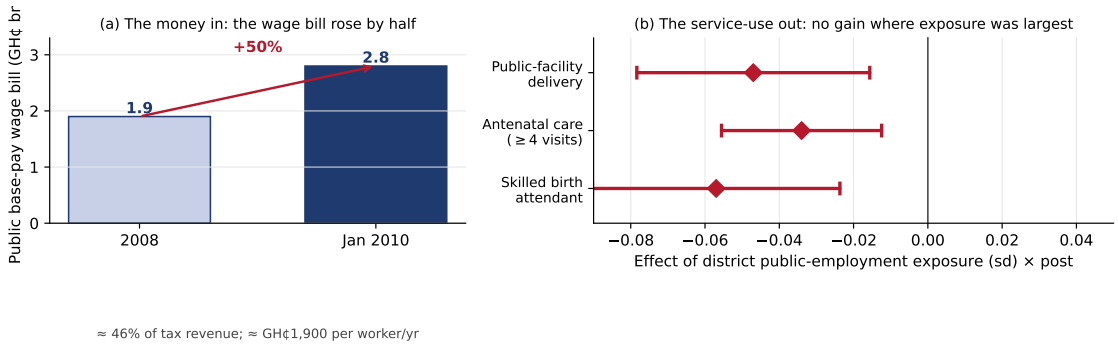
Notes: Panel (a) reports exposure-by-year coefficients for each place of delivery, with 2008 the omitted reference; panel (b) reports the pooled exposure-by-post coefficients, which sum to zero by construction. Bars are 95 percent confidence intervals from standard errors clustered by district.

the public services those workers staff, and where the public payroll absorbed most it pushed births out of the health system altogether. The reform looks more like a transfer to public-sector insiders than an investment that reached the services households use. Because exposure is concentrated in Ghana's larger towns, the estimates describe urban Ghana rather than the country as a whole.

The magnitudes make the trade-off concrete. Cavalcanti (2009) estimates that the spine raised the base-pay wage bill from GH¢ 1.9 billion in 2008 to GH¢ 2.8 billion by January 2010, a roughly fifty-percent increase and, as it covers base pay alone, a lower bound on the total. Spread over the Cavalcanti count of about 477,000 covered employees, the increment averages on the order of GH¢ 1,900 per worker per year—in a country where the public wage bill already absorbed some 46 percent of all tax revenue and 11 percent of GDP before the reform. That is the scale of the resource the spine redirected to salaries; the same outlay routed to the non-wage inputs of public clinics and schools—drugs, fuel, maintenance, supervision—would have been a large sum. The downstream estimates say that the cedis spent on the raise bought no measurable increase in the public services households actually used. A back-of-the-envelope cost per additional public-facility delivery is therefore not even well defined: the denominator, the extra utilisation the raise generated, is indistinguishable from zero. Figure 8 sets the two sides side by side.

A natural worry is that pulling deliveries away from public facilities, or squeezing

Figure 8: The Fiscal Injection and Its Return in Public-Service Use



Notes: Panel (a) plots the public base-pay wage bill before the reform (2008) and at implementation (January 2010), in GH¢ billion, from Cavalcanti (2009); the pre-reform wage bill was about 46 percent of tax revenue, and the increment averages roughly GH¢ 1,900 per covered worker per year. Panel (b) plots the district-level effect of a one-standard-deviation increase in pre-reform public-employment exposure, interacted with the post-reform indicator, on three measures of public-service use, with 95 percent confidence intervals (Table 4); standard errors clustered by district. A large fiscal injection coincides with no measurable gain—if anything a relative decline—in citizens' use of public services where the public sector was largest.

the clinics themselves, harmed children. It does not appear to. Table 20 applies the same design to the children's recode. Under-five mortality is flat with respect to exposure—no sign that the reallocation cost lives—and child stunting is statistically unchanged. The one margin that moves is measles vaccination, a service delivered almost entirely through public clinics: like facility delivery and antenatal care, it rose slightly less where the public sector was largest. The child-health evidence thus splits along the same line as the utilisation results. Services delivered through the public system track public-sector intensity weakly downward after the reform, while survival, the outcome that would register real harm, does not move—the pattern one expects if households substituted toward private care rather than going without (Table 20, in the appendix).

8 Conclusion

A government that wants better schools and clinics has an obvious lever: pay the teachers, nurses, and civil servants who run them more. Ghana pulled that lever hard. The 2010 Single Spine Pay Policy placed roughly 400,000 public workers on a single salary spine and raised the public base-pay bill by about half in two years, and measured against formal private workers it raised public earnings. Yet the money did not go where the reform's own billing promised. It bypassed the low-paid the spine was sold to help, accruing instead to workers with secondary schooling and above and to

women, and steepening a public return to schooling that was already twice the private one. And it did not reach the services those workers deliver: in the districts where the public sector loomed largest, citizens' use of public clinics rose by less, not more, through 2022, antenatal care and skilled attendance fell behind with it, and the births displaced from public facilities were delivered at home rather than in private clinics. A reform marketed as lifting the floor and strengthening the state operated, in its incidence, as a transfer to public-sector insiders.

That verdict is only visible once the comparison group is chosen with care, and this is the paper's most portable lesson. Against all non-public workers—the conventional choice, and the one in the closest prior study—the reform looks like nothing happened, because informal nominal wages raced ahead over the commodity boom and dragged the counterfactual up with them. Against formal private workers, who share the public sector's institutional and macroeconomic environment, the same reform plainly raised pay. In economies where most of the private sector is informal and nominal wages track global prices—much of the developing world—the measured public-sector pay premium turns on this single choice. A difference-in-differences is only as credible as the trend its control group would have followed, and informal earnings during a boom are the wrong trend.

Two lessons follow for the design of public pay. First, the incidence of a pay reform is a property of its grade structure, not its average size. A spine that raises the floor but climbs steeply across grades, laid over old scales that had eroded most at the top, delivers its real gains to the educated rather than the bottom—so a reform can be egalitarian in its billing and widen the very gaps, by skill and by grade, among the workers it covers. Second, higher salaries are not self-executing improvements in public services. Pay shapes who applies, who stays and how hard they work; carried one step further, to the services households actually receive, the link does not hold on its own. Paying public workers more is first of all a transfer to public workers, and in Ghana over the decade after the spine it went no further.

Governments that spend a large share of scarce revenue on public pay—Ghana's wage bill already absorbed close to half of tax receipts before the reform—might therefore ask a sharper question than how much to pay. What decides whether the money works is who sits where on the scale, and what, besides salaries, a clinic or a classroom needs to function.

Limitations. What the design settles and what it leaves open is worth stating plainly. Cohort cells defined on birth year, gender and ethnicity cannot be entered or left in response to the policy, so the estimates are free of selection on the characteristics a worker can choose, and the incidence result is stable across every treatment of composition reported here. What the cells cannot remove is time-varying selection within them. A specification that holds the public workforce at its pre-reform distribution of schooling brings the pooled effect close to zero—which is what the framework predicts, since schooling is the margin through which the spine operated—but whether the pooled gain is smaller than the benchmark implies cannot be settled with these data. What would settle it is individual administrative payroll records, which are not currently available for research use.

The earnings analysis covers the formal sector—about a fifth of Ghanaian employment—and rests on a single reform in one country; the lesson that travels furthest is the methodological one about the comparison group, which applies wherever a large, fast-growing informal sector sits between public and formal-private pay.

Extensions. Two questions follow directly from these estimates. The first is general equilibrium. Ghana's public wage bill rose by roughly half in a single year, and where that money went—into private wages through competition for skilled labour, into the size of the public workforce, or into the deficit that financed it—is a question these estimates frame but do not answer. It is also the question that decides whether a reform on this scale is affordable elsewhere. The design extends to it: the cohort cells that identify the pay effect can be followed into private-sector earnings, and the wage-bill series is public.

The second is measurement. Ghana's public payroll follows the same employees across the migration onto the spine, and the Fair Wages and Salaries Commission, which administered that migration, holds the grade placements themselves. With access to those records, the schedule in equation (1) could be estimated in its own units and the erosion wedge recovered grade by grade rather than inferred from schooling; following the same employees across the migration would separate a change in pay from a change in who holds the job; and records of hires, transfers and promotions would test the sorting channel directly rather than by inference from the education gradient.

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Online Appendix

For Online Publication

A The All-Private Control Group

This appendix applies the design of Ampofo and Tchatoka (2019) in these data, using their broad all-private control group and nominal earnings, to establish that the control-group divergence in Figure 4 reflects the choice of comparison group rather than a coding difference. Table 6 reports the pre-policy descriptives for the A&T sample, and Table 7 reports their headline exhibit—the quantile difference-in-differences of the reform on log earnings—with their published estimates listed in the note.

Under the all-private control the reform is undetectable in these data, which is the conclusion A&T reach. Their published magnitudes are not recovered. They report an OLS interaction of 2.107 and a tenth-quantile interaction of 3.286, against -0.134 and -0.468 here. The divergence extends to the level term, which they report as negative at every quantile and which is positive throughout in these data, and to the estimation sample: 24,639 observations against 31,398 under the same stated restrictions. Coefficients of the magnitude they report imply earnings gains of several hundred percent. The contrast that Figure 4 rests on is therefore between control groups within these data.

Table 6: Pre-Policy Descriptive Statistics (GLSS Waves 4+5, 1999+2005)

	(1) Public Sector		(2) All Private		(3) Formal Private		(4) Full Sample	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Log monthly earnings, nominal GH₵(A&T scale)	4.714	0.792	3.638	1.219	4.384	0.878	3.751	1.226
Log weekly hours	3.767	0.423	3.647	0.597	3.913	0.439	3.659	0.583
Imputed years of schooling	13.094	2.641	9.798	2.551	11.758	2.518	10.224	2.792
Potential work experience (years)	23.564	10.512	23.171	10.911	21.103	11.038	23.222	10.860
1 = married or cohabiting	0.171	0.377	0.186	0.389	0.141	0.349	0.184	0.388
1 = female	0.151	0.358	0.275	0.446	0.123	0.329	0.262	0.440
1 = trade union member	0.701	0.458	0.060	0.237	0.433	0.496	0.127	0.333
1 = household head	1.000	0.017	0.999	0.029	0.999	0.027	0.999	0.028
1 = father in formal/govt employment	0.093	0.290	0.049	0.216	0.091	0.287	0.054	0.225

Notes: Sample: ALL employed workers age 15-64, waves 4+5 (pre-SSPP). Earnings: log nominal monthly GH*c* (ln(earn_month)). Public sector = civil service + parastatal. Formal private = employer type 7. All private = all non-public employed workers (A&T main control group).

Table 7: Panel I: Pooled QR without Individual Controls — Log Earnings

	(1) OLS	(2) Q10	(3) Q25	(4) Q50	(5) Q75	(6) Q90
Public × Post-SSPP	-0.134	-0.468***	-0.0751	0.0697*	0.0746*	0.0652
	(0.0801)	(0.0878)	(0.0509)	(0.0360)	(0.0423)	(0.0572)
1 = Public sector	1.137***	1.279***	1.154***	0.916***	0.570***	0.315***
	(0.111)	(0.0468)	(0.0345)	(0.0243)	(0.0313)	(0.0420)
Observations	31,398	31,353	31,353	31,353	31,353	31,353

Notes: This table applies the design of Ampofo and Tchatoka (2019) (their Table 2, Panel I) to these data: log nominal monthly earnings, the all-private control group, GLSS Waves 4–6, with year, region, and industry fixed effects and bootstrap standard errors. Under their control group the reform is undetectable here—the OLS estimate is -0.134 with a standard error of 0.080—which is the conclusion they draw. Their published magnitudes are not recovered. A&T report Public × Policy of 2.107 (OLS) and 3.286 (Q10), against -0.134 and -0.468 here; they also report a public-sector level term that is negative at every quantile, where the public premium in these data is positive throughout, and their estimation sample contains 24,639 observations against 31,398 here under the same stated restrictions. Coefficients of the magnitude they report imply earnings gains of several hundred percent. The comparison relevant to Figure 4 is therefore between control groups within these data, not between these estimates and their published ones.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B Event-Study Coefficients

Table 8 reports the coefficients underlying the earnings event study of Figure 12 in the main text.

Table 8: Event Study: Public–Private Earnings Gap by Survey Year (Full Sample)

	(1) Regional exposure	(2) Difference-in-differences
Public/High-exposure × 1998	0.076	0.087
	(0.175)	(0.098)
Public/High-exposure × 1999	0.056	0.207
	(0.201)	(0.126)
Public/High-exposure × 2005	0.135*	0.018
	(0.070)	(0.070)
Public/High-exposure × 2006 (reference)	0	0
Public/High-exposure × 2013	0.002	0.254**
	(0.156)	(0.126)
Public/High-exposure × 2016	0.454**	0.251**
	(0.187)	(0.124)
Public/High-exposure × 2017	0.463**	0.292***
	(0.182)	(0.085)
Observations	36,620	9,068

Notes: Outcome: log real monthly earnings (2017 GH¢, winsorised 1–99 within wave). Survey year on the row dimension; 2006 is the omitted reference. Column (1) interacts above-median regional public-employment exposure (Wave 5) with survey year on all workers; column (2) interacts public-sector status with survey year on public and formal private workers. Controls: female, age, marital status, household headship, father in formal employment; region and survey-year fixed effects.

Standard errors clustered by region (10 clusters) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Distributional Estimates

This appendix collects the distributional estimates summarised in Section 6.3. Table 9 reports the full set of quantile difference-in-differences coefficients for log real monthly earnings, by quantile, with and without controls; it is the source for Figure 14 in the main text.

Table 9: Quantile Treatment Effects: Log Real Earnings

	Panel I: No Controls					Panel II: With Controls				
	(1) Q10	(2) Q25	(3) Q50	(4) Q75	(5) Q90	(6) Q10	(7) Q25	(8) Q50	(9) Q75	(10) Q90
Public × Post	-0.174	0.195***	0.260***	0.065	-0.051	0.035	0.110*	0.088	0.052	-0.043
	(0.108)	(0.070)	(0.052)	(0.073)	(0.088)	(0.081)	(0.056)	(0.054)	(0.060)	(0.077)
Public sector	0.693***	0.670***	0.540***	0.424***	0.336***	0.372***	0.374***	0.274***	0.122***	0.081
	(0.075)	(0.041)	(0.040)	(0.056)	(0.072)	(0.054)	(0.045)	(0.046)	(0.045)	(0.057)
Observations	5,992	5,992	5,992	5,992	5,992	5,245	5,245	5,245	5,245	5,245

Notes: Sample: public + formal private, W4–W7. Outcome: log real monthly earnings, 2017 GH¢. Controls (Panel II): education, experience (sq.), married, hhhead, fa_emp. Wave + region + industry FE as dummies. Bootstrap SE: 200 reps.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10 reports the companion within-sample quantile regression that traces the public earnings gain across the distribution relative to the formal-private control.

Table 10: The Earnings Effect Across the Distribution

	(1) (1) OLS	(2) (2) Q10	(3) (3) Q25	(4) (4) Q50	(5) (5) Q75	(6) (6) Q90
Public × Post	0.115**	0.212***	0.369***	0.313***	0.256***	0.170***
	(0.036)	(0.050)	(0.031)	(0.028)	(0.031)	(0.048)
Public sector	0.202***	0.234***	0.099***	0.029	-0.046	-0.140***
	(0.021)	(0.046)	(0.028)	(0.029)	(0.034)	(0.050)
Estimator	OLS (mean)	Q10	Q25	Q50	Q75	Q90
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,206	8,206	8,206	8,206	8,206	8,206

Notes: This table reports the public–private earnings effect of the reform across the conditional distribution of log real monthly earnings (2017 GH¢). Column 1 is the OLS (mean) difference-in-differences; columns 2–6 are quantile difference-in-differences at the 10th, 25th, 50th, 75th, and 90th percentiles. The coefficient of interest is Public × Post. A pattern of coefficients rising across quantiles would indicate that the gap widened toward the top of the distribution; a falling pattern would indicate compression relative to the formal-private control. All columns include the control vector and wave and region fixed effects; the OLS column uses region-clustered standard errors and the quantile columns use bootstrap standard errors, reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The remaining tables disaggregate the distributional effect by margin. Table 11 repeats the quantile exercise for log weekly hours, and Tables 12 and 13 report the

earnings quantile estimates separately for men and women.

Table 11: Quantile Treatment Effects: Log Weekly Hours

	(1) Q10	(2) Q25	(3) Q50	(4) Q75	(5) Q90
Public × Post	-0.055	0.021	-0.021	0.016	0.054
	(0.058)	(0.024)	(0.020)	(0.034)	(0.039)
Public sector	-0.046	-0.095***	-0.101***	-0.169***	-0.135***
	(0.046)	(0.019)	(0.017)	(0.030)	(0.036)
Observations	5,013	5,013	5,013	5,013	5,013

Notes: Sample and controls as in earnings QTE table. Outcome: log weekly hours.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12: QTE Earnings: Male workers

	(1) Q10	(2) Q25	(3) Q50	(4) Q75	(5) Q90
Public × Post	0.145	0.121*	0.056	0.001	-0.108
	(0.099)	(0.067)	(0.059)	(0.066)	(0.099)
Observations	3,708	3,708	3,708	3,708	3,708

Notes: Sample restricted to male workers. Controls and FE as above.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13: QTE Earnings: Female workers

	(1) Q10	(2) Q25	(3) Q50	(4) Q75	(5) Q90
Public × Post	0.051	0.134	0.208*	0.432***	0.247*
	(0.149)	(0.097)	(0.117)	(0.122)	(0.137)
Observations	1,537	1,537	1,537	1,537	1,537

Notes: Sample restricted to female workers. Controls and FE as above.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

D Additional Robustness

Selection on unobservables (Oster bound). Table 14 reports the selection-on-unobservables bound of Oster (2019) for the main four-wave difference-in-differences. At Oster's recommended benchmark for the maximum R^2, the degree of selection on unobservables relative to observables that would drive the coefficient to zero is about 1.8, and the bias-adjusted estimate remains positive; the result is overturned only under the extreme

and, for log earnings, implausible assumption that treatment and covariates leave no idiosyncratic variation in pay. A complementary check that lets the returns to education and experience vary freely across waves leaves the estimate intact (about 0.17).

Table 14: Sensitivity to Selection on Unobservables (Oster Bounds)

	Value
Controlled estimate $\hat{\beta}$ (main spec)	0.115***
	(0.036)
Uncontrolled estimate $\mathring{\beta}$	0.211
Controlled $\tilde{R}^2$	0.315
Uncontrolled $\mathring{R}^2$	0.171
At Oster's recommended $R^2_{\max} = 1.3\,\tilde{R}^2$:	
Bias-adjusted $\beta^*(\delta = 1)$	0.053
δ for $\beta^* = 0$	1.84
At the extreme $R^2_{\max} = 1$:	
Bias-adjusted $\beta^*(\delta = 1)$	-0.340
δ for $\beta^* = 0$	0.25
Flexible-price check ($\times$ wave)	0.169***
	(0.047)
Observations	8,206

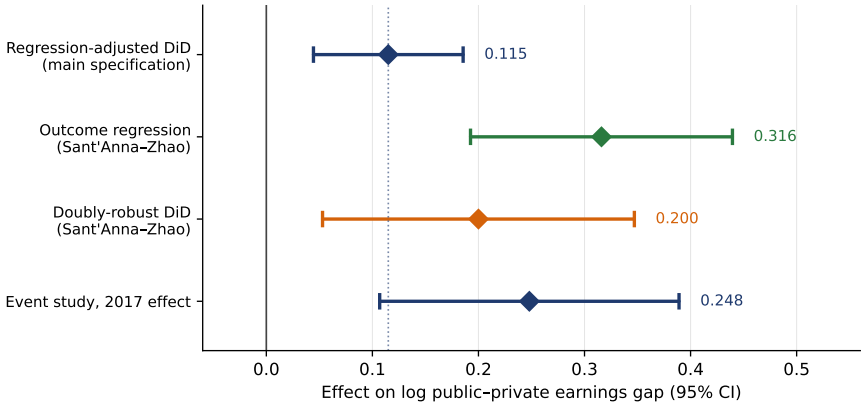
Notes: Selection-on-unobservables bounds following Oster (2019) for the main four-wave difference-in-differences (Table 2, column 3). $\hat{\beta}$ and $\tilde{R}^2$ are the coefficient and R^2 of the controlled regression; $\mathring{\beta}$ and $\mathring{R}^2$ are those of a regression with only the treatment, public-sector indicator, and wave and region fixed effects. δ is the degree of selection on unobservables, relative to selection on the included observables, that would drive the coefficient to zero; $\delta > 1$ indicates the estimate is robust. $\beta^*(\delta = 1)$ is the coefficient adjusted for selection on unobservables as strong as on observables. The flexible-price check lets the returns to education and experience vary by wave. Standard errors clustered by region.

Doubly-robust estimation. Figure 9 places the doubly-robust repeated-cross-section estimate of Sant'Anna and Zhao (2020) alongside the regression-adjusted, outcome-regression and long-run event-study estimates. All are positive and significant, and the doubly-robust estimate sits among them.

Sensitivity to parallel-trends violations. Table 15 reports the relative-magnitudes sensitivity analysis of Rambachan and Roth (2023): the long-run effect remains significant for post-reform deviations from parallel trends up to about three-quarters of the pre-reform deviation.

Continuous regional-exposure design. Table 16 re-estimates the regional-exposure event study with the continuous standardised public-employment share in place of the median split, and with a leave-one-out version of the share. The growing post-reform

Figure 9: The Earnings Effect Across Estimators



Notes: Point estimates and 95% confidence intervals for the effect of the reform on the log public–private earnings gap, by estimator. The regression-adjusted difference-in-differences is the pooled four-wave specification; the outcome-regression and doubly-robust estimates are those of Sant'Anna and Zhao (2020); the event-study estimate is the 2017 coefficient. Standard errors clustered by region.

Table 15: Sensitivity to Parallel-Trends Violations (HonestDiD)

$\bar{M}$	Lower bound	Upper bound
Original (point identified)	0.108	0.388
0.50	0.051	0.577
0.75	0.023	0.659
1.00	−0.096	0.790
1.50	−0.308	1.007
2.00	−0.522	1.223

Notes: Robust 95% confidence sets for the long-run (2017) event-study effect under the relative-magnitudes restriction of Rambachan and Roth (2023), which bounds any post-reform deviation from parallel trends at $\bar{M}$ times the deviation observed in the pre-reform period (1999). The point-identified estimate is 0.248 (standard error 0.072). The interval first includes zero between $\bar{M} = 0.7$ and $\bar{M} = 0.8$, so the effect survives post-reform violations up to about three-quarters the size of the pre-reform deviation—which is itself small and statistically insignificant (0.069, standard error 0.091). With a single pre-period the relative-magnitudes restriction is necessarily weak; the doubly-robust estimate and the Oster (2019) bound (Table 14) are the stronger guards against composition and selection.

pattern and flat pre-trends are unchanged, and the capital (Greater Accra) is a low-exposure region, so the design is not driven by it.

Regional subsamples and alternative clustering. The four-wave difference-in-differences is significant on the full sample and essentially unchanged when Greater Accra is excluded; bootstrap and two-way (wave-by-region) clustering leave the estimate significant.

Non-wage benefits. Public workers hold more generous benefits in levels, but the reform did not shift benefit coverage differentially: the difference-in-differences on pen-

Table 16: Continuous Regional-Exposure Design (Shift-Share Diagnostics)

	(1) Continuous dose	(2) Leave-one-out
Exposure (s.d.) × 1998	0.006	0.005
	(0.139)	(0.139)
Exposure (s.d.) × 1999	-0.167	-0.169
	(0.180)	(0.181)
Exposure (s.d.) × 2005	0.004	0.000
	(0.073)	(0.073)
Exposure (s.d.) × 2006 (reference)	0	0
Exposure (s.d.) × 2013	0.101	0.099
	(0.116)	(0.117)
Exposure (s.d.) × 2016	0.282***	0.281***
	(0.081)	(0.080)
Exposure (s.d.) × 2017	0.308***	0.307***
	(0.098)	(0.097)
Observations	36,620	36,620

Notes: Outcome: log real monthly earnings. Exposure is the standardised Wave 5 regional public-employment share interacted with survey year (2006 omitted); a one-standard-deviation increase in regional exposure corresponds to roughly 0.16 in the share. Column (1) uses the raw regional share; column (2) recomputes each Wave 5 respondent's share excluding their own record. Both use all workers with region and survey-year fixed effects and SE clustered by region. Greater Accra is a low-exposure region, so the design is not driven by the capital.

Standard errors clustered by region (10 clusters) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: Regional Subsamples and Alternative Clustering

	(1) (1) All regions	(2) (2) Excl. Accra	(3) (3) Bootstrap SE	(4) (4) Two-way
Public × Post	0.115**	0.111**	0.115***	0.115***
	(0.036)	(0.046)	(0.034)	(0.015)
Public sector	0.202***	0.207***	0.202***	0.202**
	(0.021)	(0.037)	(0.030)	(0.035)
Sample	All regions	Excl. Accra	All regions	All regions
SE clustering	Region	Region	Bootstrap (50 reps)	Wave × Region
Observations	8,206	6,217	8,206	8,206

Notes: Outcome: log real monthly earnings (2017 GH₵), Waves 4–7. Column (2) excludes Greater Accra. Column (3) uses bootstrap SE (50 reps); column (4) two-way clusters by wave and region. Wave and region fixed effects absorbed; SE clustered by region unless noted.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

sion, paid holiday, paid sick leave, and medical-care coverage is insignificant (medical care is marginally negative), so the earnings results are not offset by a measured change in non-wage compensation.

Table 18: Non-Wage Benefit Proxies: SSPP DiD Estimates

	(1) 1 = pension/social security
Public × Post	0.057
	(0.049)
Observations	8,198

Notes: Each column is a separate OLS DiD regression. Outcome: binary indicator for benefit take-up.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

E Service-Delivery Extension: Robustness and Child Health

This appendix collects the supporting estimates for the service-delivery results of Section 7. Table 19 shows that the public-facility-delivery and antenatal-care effects survive a full set of urban-status-by-year interactions and region-by-year fixed effects, and that they are a feature of urban rather than rural clusters. Table 20 applies the same district-exposure design to the children's recode: under-five mortality and stunting are unmoved, while measles vaccination—a service delivered through public clinics—tracks the utilisation results.

Table 19: Service Delivery: Robustness of the District Difference-in-Differences

	Public-facility delivery	ANC ($\geq$4) visits	Skilled birth attendant
Baseline (district + year FE)	-0.047***	-0.034***	-0.057***
	(0.016)	(0.011)	(0.017)
+ urban × year	-0.033**	-0.026**	-0.049***
	(0.015)	(0.010)	(0.016)
Region × year FE	-0.037**	-0.026***	-0.035**
	(0.018)	(0.009)	(0.014)
Rural sample only	0.029	0.006	0.024
	(0.022)	(0.017)	(0.024)

Notes: Each cell reports the exposure (sd) × post coefficient from a separate difference-in-differences (as in Table 4), under the specification named in the row. *Baseline* includes district and survey-year fixed effects; the second row adds a full set of urban-status × survey-year interactions; the third replaces survey-year effects with region × survey-year effects (identifying the coefficient from variation across districts within a region and year); the fourth restricts to rural clusters. All are weighted by DHS sampling weights.

Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 20: Public-Employment Exposure and Child Health

	Stunted (HAZ < -2)	Measles vaccinated	Under-5 mortality
Exposure (sd) × Post	0.016	-0.013*	0.001
	(0.012)	(0.008)	(0.003)
Pre-reform mean	0.277	0.869	0.075
Post-reform mean	0.186	0.895	0.038
Observations	9,426	12,875	21,920
DHS rounds	3	4	4
District FE	Yes	Yes	Yes
Survey-year FE	Yes	Yes	Yes
Districts (clusters)	102	102	102

Notes: Difference-in-differences on the DHS children's recode (children born in the five years before each survey), pre-reform 2003 and 2008 versus post-reform 2014 and 2022. *Exposure* is the standardised pre-reform (2000 Census) public-employment share of the child's district, assigned from cluster GPS. Stunting is height-for-age below -2 SD on the WHO standard (available from 2008, so that column uses three rounds); measles vaccination is defined for children aged 12–59 months; under-5 mortality is an indicator that the child was not alive at interview. All columns include district and survey-year fixed effects and DHS sampling weights.

Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Supplementary Tables and Figures

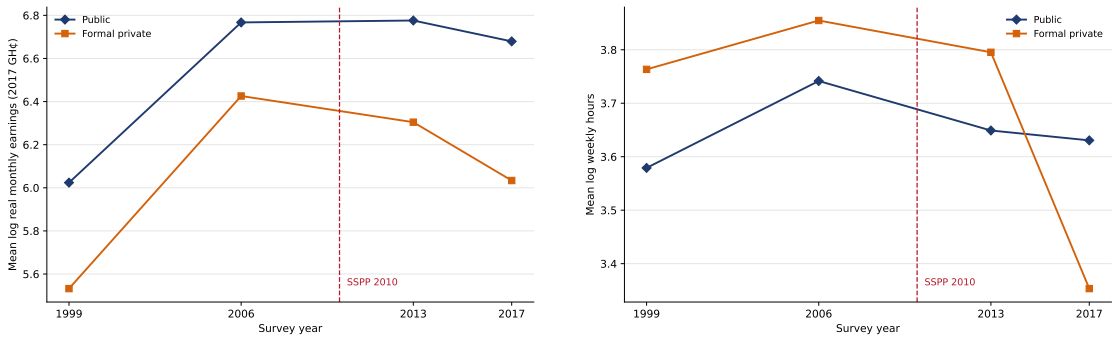
Table 21: OLS Difference-in-Differences: Earnings and Hours

	Log Real Earnings			Log Weekly Hours	
	(1)	(2)	(3)	(4)	(5)
Public × Post	0.131**	0.155***	0.115**	0.060	0.058
	(0.051)	(0.041)	(0.036)	(0.040)	(0.045)
Public sector	0.501***	0.498***	0.202***	-0.136**	-0.140**
	(0.055)	(0.032)	(0.021)	(0.045)	(0.059)
Fixed effects	None	Wave, region	Wave, region	Wave, region	Wave, region
Individual controls	No	No	Yes	No	Yes
Observations	9,699	9,699	8,206	9,113	7,708

Notes: This table reports pooled OLS difference-in-differences estimates comparing public-sector workers to formal private (written-contract) workers over the four-wave window (1999–2017), the same window as the event study. The coefficient Public × Post is the change in the public–private gap after the 2010 reform. The dependent variable is log real monthly earnings (2017 GH¢, winsorised 1–99 within wave) in columns 1–3 and log weekly hours in columns 4–5. The earnings columns build up from a raw two-by-two difference-in-differences with no fixed effects or controls (column 1), adding wave and region fixed effects (column 2), and the individual control vector (column 3, the main specification). The control vector is education, experience and its square, marital status, household headship, and father's formal employment; union membership is excluded because it is unrecorded in the 2016/17 wave. Standard errors, clustered by region, are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 10: Log Real Earnings and Hours by Sector, 1999–2017



(a) Log real monthly earnings

(b) Log weekly hours

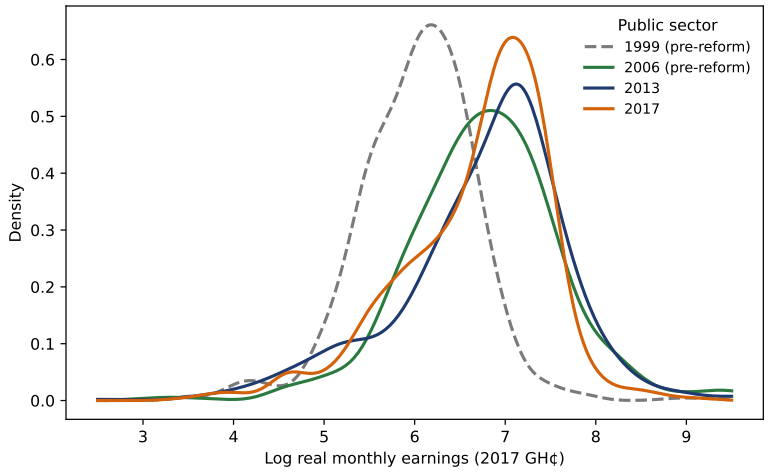
Notes: Sample means for public versus formal private workers across the four GLSS waves. Earnings deflated to 2017 GH¢. The vertical dashed line marks SSPP implementation (January 2010).

Table 22: Composition of the Public and Formal-Private Samples by Wave

	1999	2006	2013	2017
Panel A. Public sector				
Years of schooling	12.2	13.3	13.7	14.7
Potential experience (yrs)	23.9	22.5	19.1	17.7
Female (%)	28	29	39	37
Union member (%)	76	72	71	n.a.
Household head (%)	82	79	71	70
Father in formal emp. (%)	37	10	11	18
Observations	651	772	1377	1386
Panel B. Formal private				
Years of schooling	10.4	11.8	12.1	11.1
Potential experience (yrs)	18.4	19.1	18.0	16.0
Female (%)	23	25	27	29
Union member (%)	22	42	29	n.a.
Household head (%)	75	72	70	62
Father in formal emp. (%)	39	8	12	10
Observations	645	501	1044	3546

Notes: Each cell is a sample mean for the indicated GLSS wave, restricted to employed workers aged 15–64 with positive earnings in the public or formal-private sector—the estimation sample. Schooling and experience are in years; the remaining rows are percentages. Stable means across waves indicate limited recomposition of either sector over the study period. "n.a." marks a characteristic not collected in that wave (union membership is unavailable in GLSS 7). Source: GLSS Waves 4–7.

Figure 11: Distribution of Log Real Public Earnings, Before and After the Reform



Notes: Kernel densities of log real monthly earnings (2017 GH¢) for public-sector workers in the pre-reform wave (2006) and the two post-reform waves (2013, 2017). The rightward shift, concentrated in the lower half of the distribution, is the descriptive analogue of the spine raising base pay proportionally most at the lower grades. Source: GLSS Waves 5–7.

Table 23: Did Composition Move? Difference-in-Differences on Worker Characteristics

	Public × Post	Public mean pre-reform	Public mean post-reform
	(1)	(2)	(3)
Years of schooling	1.201***	12.820	14.228
	(0.220)		
Potential experience	-2.257***	23.285	18.872
	(0.469)		
Female	0.043	0.298	0.379
	(0.026)		
Married	-0.057**	0.163	0.082
	(0.021)		
Household head	0.023	0.810	0.747
	(0.023)		
Father in formal employment	0.056**	0.227	0.148
	(0.020)		

Notes: Each row reports the Public × Post coefficient from the main difference-in-differences specification estimated with the named worker characteristic as the dependent variable, on the estimation sample of Table 2. All regressions include survey-wave and region fixed effects. A coefficient indistinguishable from zero indicates that the observable composition of public employment did not shift differentially relative to formal private employment after the reform; a significant coefficient identifies the margin along which it did. Columns 3 and 4 report the raw pre- and post-reform means among public employees.

Standard errors clustered by region. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

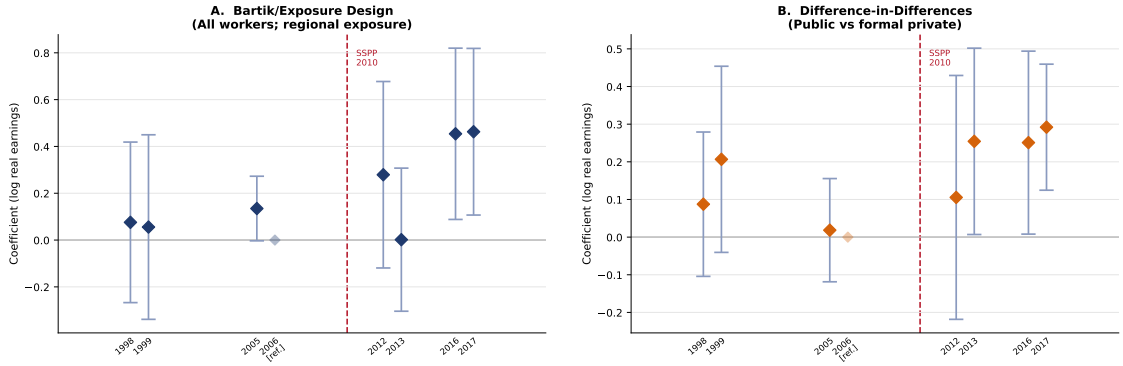
Table 24: Persistence of the Effect Across Horizons

	(1)	(2)	(3)	(4)	(5)
	(1) Earn, SR	(2) Earn, MR	(3) Earn, LR	(4) Hrs, SR	(5) Hrs, MR
Public × Post	0.177**	0.115**	0.112***	0.006	0.058
	(0.057)	(0.036)	(0.028)	(0.056)	(0.045)
Public sector	0.144***	0.202***	0.229***	-0.133*	-0.140**
	(0.031)	(0.021)	(0.020)	(0.064)	(0.059)
Sample	W4–6	W4–7	W4–5, W7	W4–6	W4–7
Period	Short-run	Medium-run	Long-run (7 yrs)	Short-run	Medium-run
Observations	4,274	8,206	6,102	4,124	7,708

Notes: This table traces the public–private effect of the reform across horizons under the formal-private control, with the pre-reform period pooling the 1999 and 2005 waves. Columns 1–3 report log real monthly earnings (2017 GH¢): a short horizon through 2013, the full horizon through 2017, and a long difference comparing the pre-reform waves with 2017 alone; columns 4–5 report log weekly hours over the short and full horizons. The coefficient of interest is Public × Post. All columns include the control vector (education, experience and its square, marital status, household headship, and father's formal employment) and wave and region fixed effects. Standard errors, clustered by region, are in parentheses.

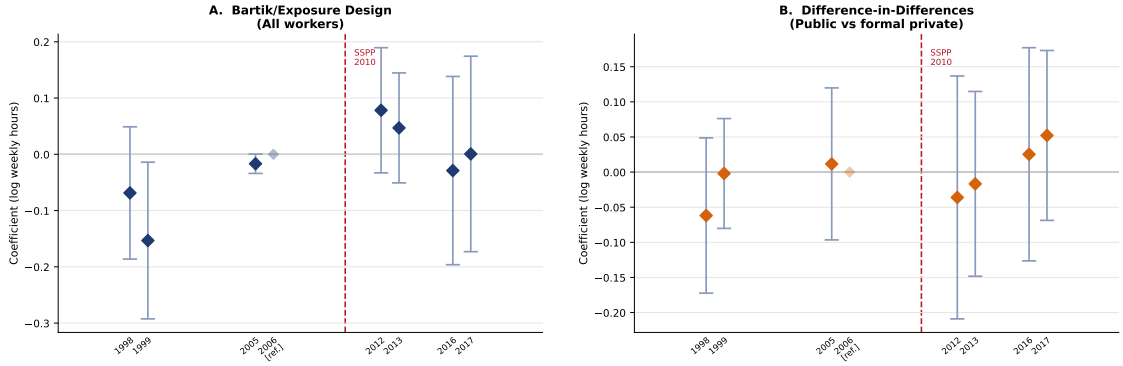
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 12: Event Study: Regional-Exposure (Panel A) and Difference-in-Differences (Panel B)



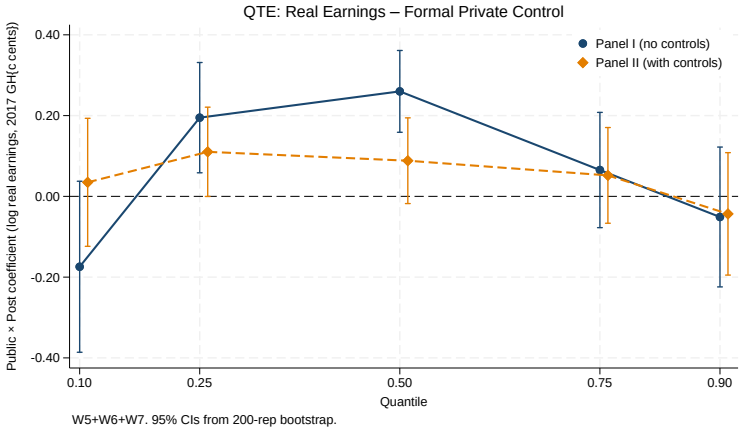
Notes: Outcome is log real monthly earnings (2017 GH¢, winsorised 1–99 within wave). The horizontal axis is the actual GLSS interview year; 2006 is the omitted reference. *Panel A* (exposure): high-exposure regions have a pre-reform public employment share ≥ 0.383; $N \approx 36{,}620$. *Panel B* (DiD): public versus formal private workers; $N \approx 9{,}068$. Controls: female, age, marital status, household headship, and father in formal employment; region and survey-year effects; standard errors clustered by region (ten clusters). The dashed line marks the SSPP (January 2010).

Figure 13: Event Study: Log Weekly Hours



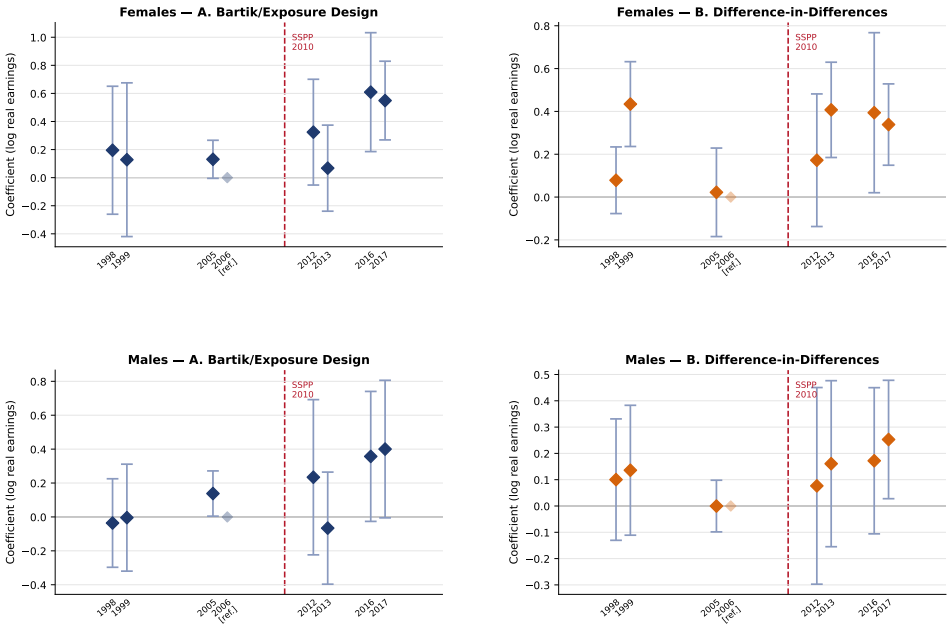
Notes: Event study for log weekly hours, regional-exposure (left) and difference-in-differences (right) designs, with 2006 the omitted reference. Controls and clustering as in Figure 12.

Figure 14: Quantile Difference-in-Differences: Log Real Monthly Earnings



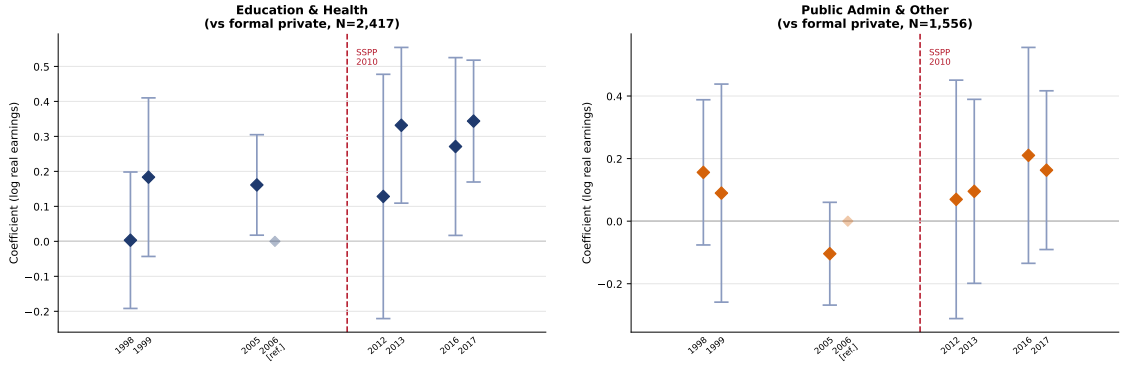
Notes: Quantile DiD coefficients on Public × Post (formal private control), with and without controls, with 95% bootstrap intervals (200 replications). Outcome: log real monthly earnings. Table 9 reports the underlying estimates.

Figure 15: Event Study by Gender: Log Real Monthly Earnings



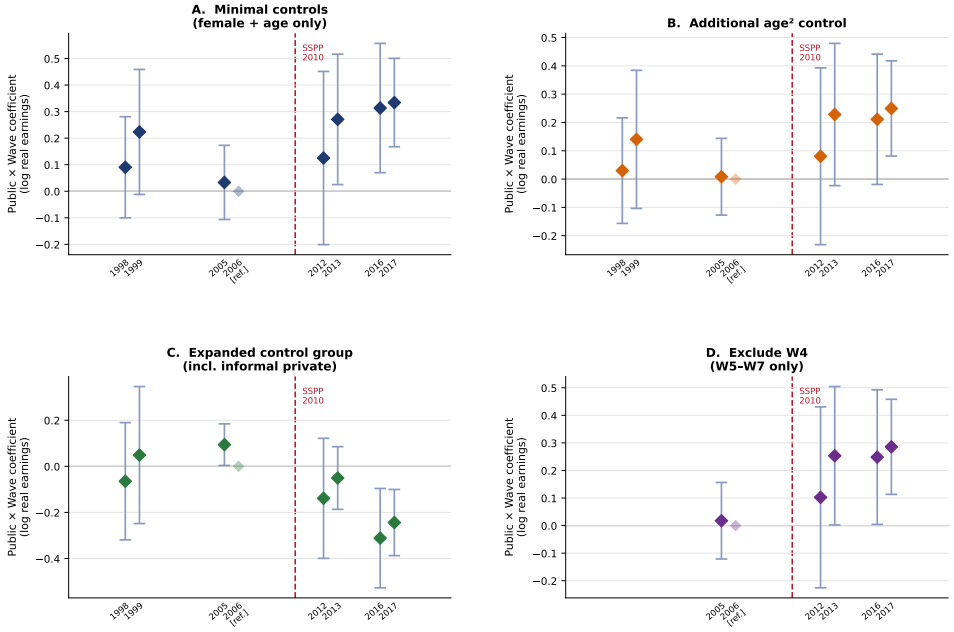
Notes: Event study estimated separately for women (top) and men (bottom), exposure (left) and DiD (right). The female DiD carries a significant 1999 pre-reform coefficient, so the female post-reform estimates are not cleanly identified. Controls and clustering as in Figure 12.

Figure 16: Event Study by Public Sub-Sector: Log Real Monthly Earnings



Notes: Education & Health versus Public Administration & Other, each against the formal-private control. Reference year 2006. Controls and clustering as in Figure 12.

Figure 17: Sensitivity of the DiD Event Study



Notes: Panel A minimal controls; Panel B adds age squared; Panel C expands the control group to include informal private workers; Panel D excludes Wave 4. Reference year 2006; standard errors clustered by region.

Table 25: Wave-Range and Earnings-Measure Sensitivity

	(1) (a) All Private	(2) (b) Formal Only	(3) (c) Non-Agric Private
Public × Post-SSPP	-0.0269	0.152**	0.130**
	(0.0584)	(0.0664)	(0.0494)
1 = Public sector	0.359***	0.0606	0.114**
	(0.0527)	(0.0525)	(0.0396)
Observations	23,390	4,573	16,955

Notes: All columns use the formal-private control group; the all-private (A&T) comparison is in Appendix A and Fig. 17, Panel C. Column (1): nominal earnings, W4–6. Column (3): no individual controls. Column (4): preferred specification. Column (5): W4–6 window. Wave and region fixed effects absorbed; SE clustered by region.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 26: Heterogeneity in the Earnings Effect by Education and Sex

	(1) All	(2) Primary	(3) Second.	(4) Tertiary	(5) Men	(6) Women
Public × Post	0.115**	-0.238	-0.150**	0.328***	0.075	0.247***
	(0.036)	(0.184)	(0.051)	(0.069)	(0.064)	(0.063)
Public sector	0.202***	0.176	0.240***	0.229***	0.189***	0.294***
	(0.021)	(0.136)	(0.053)	(0.043)	(0.031)	(0.072)
Observations	8,206	823	2,836	5,329	5,773	2,433

Notes: This table reports the public–private earnings effect of the SSPP estimated separately on subsamples defined by education and sex. The dependent variable is log real monthly earnings (2017 GH¢); the regressor of interest, Public × Post, is the difference-in-differences coefficient. Column 1 uses the full public and formal-private sample (Waves 4–7). Columns 2–4 split by schooling: primary (fewer than 8 years), secondary (8–12 years), and tertiary (13 or more years). Columns 5–6 split by sex. Every column is a separate OLS regression including wave and region fixed effects and the full individual control vector (education, experience and its square, marital status, household headship, and father's formal employment). Standard errors, clustered by region, are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 27: Returns to Education by Sector and Period

	(1) (1) All	(2) (2) Public	(3) (3) Private	(4) (4) Pre	(5) (5) Post
Education years	0.092***	0.129***	0.072***	0.139***	0.087***
	(0.007)	(0.005)	(0.011)	(0.011)	(0.008)
Experience	0.055***	0.035***	0.062***	0.045***	0.057***
	(0.004)	(0.004)	(0.004)	(0.008)	(0.004)
Experience^2	-0.001***	-0.000***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Public $\times$ Post	0.105**				
	(0.035)				
Public sector	0.204***			0.191**	0.315***
	(0.023)			(0.059)	(0.057)
Sample	All formal	Public only	Private only	Pre (W4–5)	Post (W6-7)
Observations	8,988	3,966	5,022	2,296	6,692

Notes: This table reports Mincer-type earnings regressions of log real monthly earnings on years of schooling, potential experience and its square. Column 1 pools all formal workers; columns 2–3 estimate the return separately for the public and formal-private sectors; columns 4–5 re-estimate it before (1999–2005) and after (2013–2017) the reform. The coefficient on years of schooling is the return to a year of education. Columns 2–3 show that the public sector rewards a year of schooling about twice as richly as the formal private sector; columns 4–5 show the public-sector premium (the Public sector coefficient) rising across the reform. The steepening of the public education gradient itself is established by the triple-difference in Table 28. All columns include wave and region fixed effects; standard errors, clustered by region, are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 28: Skill and the Sorting Margin

	(1) (1) P(public)	(2) (2) Earnings	(3) (3) Triple	(4) (4) Union
Tertiary × Post	-0.273***			
	(0.041)			
Public × Post	0.972***	0.105**	-0.561***	-0.012
	(0.006)	(0.035)	(0.128)	(0.047)
Public sector		0.204***	0.227***	0.315***
		(0.023)	(0.026)	(0.066)
Education years		0.092***	0.078***	0.026***
		(0.007)	(0.009)	(0.003)
Pub × Post × Edu yrs			0.048***	
			(0.009)	
Outcome	P(public)	ln(earn), public	ln(earn)	Union member
Interaction	Tertiary × Post	No	DiD × Edu	No
Observations	8,988	8,988	8,988	4,609

Notes: This table reports evidence on how the reform interacted with worker skill. Column 1 is a linear probability model for public-sector employment, including a tertiary-education-by-post interaction; it is reported with caution because the regressor set is not cleanly identified for a sector-choice outcome, and no conclusion about net reallocation is drawn from it. Column 2 is the earnings difference-in-differences within the public and formal-private sample. Column 3 adds a triple interaction of Public, Post, and years of schooling: its positive coefficient means the public-sector return to education rose after the reform. Column 4 examines union membership. All columns include wave and region fixed effects; standard errors, clustered by region, are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 29: Hours and the Labour-Supply Response to the Reform

	(1) (1) Hours	(2) (2) Hours	(3) (3) Hours+ind.	(4) (4) First stage	(5) (5) 2SLS
Public × Post	0.060	0.058	0.004	0.115**	
	(0.040)	(0.045)	(0.052)	(0.036)	
Public sector	-0.136**	-0.140**	-0.110	0.202***	-0.242
	(0.045)	(0.059)	(0.063)	(0.021)	(0.138)
Log real earnings (instrumented)					0.499
					(0.411)
Specification	OLS	OLS	OLS	First stage	2SLS (Frisch)
Controls	No	Yes	Yes + industry FE	Yes	Yes
KP F-stat					11.5
Observations	9,113	7,708	5,013	8,206	7,708

Notes: This table examines whether the reform changed hours of work. Columns 1–3 report OLS difference-in-differences estimates for log weekly hours, adding the control vector in column 2 and industry fixed effects in column 3. Columns 4–5 attempt to recover a labour-supply elasticity by instrumenting log real earnings with the reform indicator (Public × Post): column 4 is the first stage and column 5 the two-stage least-squares estimate, whose coefficient on instrumented earnings is the implied Frisch elasticity. The Kleibergen–Paap first-stage F-statistic is reported at the foot of the table; at 11.5 it clears the conventional weak-instrument threshold. The implied Frisch elasticity in column 5 is nonetheless small (0.50) and imprecisely estimated, statistically indistinguishable from zero, consistent with the null OLS hours estimates in columns 1–3. All columns include wave and region fixed effects; standard errors, clustered by region, are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 30: Reweighting to the Pre-Reform Covariate Distribution

	Baseline (1)	Demographics (2)	+ Experience (3)	+ Schooling (4)
Public × Post	0.115***	0.079*	0.048	0.026
	(0.036)	(0.041)	(0.040)	(0.043)
Observations	8,206	8,206	8,206	8,206
Controls	Yes	Yes	Yes	Yes
Wave and region FE	Yes	Yes	Yes	Yes

Notes: Column 1 repeats the baseline specification. Columns 2–4 reweight post-reform observations, separately within each sector, so that the named characteristics have the same distribution as in that sector before the reform (DiNardo et al., 1996); weights come from a sector-specific logit for being observed after the reform and are trimmed at the 99th percentile. Column 2 balances pre-determined demographics (gender, marital status, household headship, father's formal employment); column 3 adds potential experience; column 4 adds years of schooling. Schooling and experience are margins along which the reform may itself have induced sorting, so column 4 is a lower bound that nets out part of the treatment response rather than a cleaner estimate. Standard errors clustered by region.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 31: Is the Formal Private Control Group Contaminated by the Commodity Boom?

	Public × Post (1)	Observations (2)
Preferred design	0.089**	8,154
	(0.043)	
Industry × wave FE	0.061	5,235
	(0.057)	
Services only	0.081	4,372
	(0.072)	
Excl. industry and construction	0.046	7,875
	(0.051)	
Excl. Western (oil)	0.125**	7,173
	(0.048)	
Excl. Western and Ashanti	0.149***	5,894
	(0.052)	
Region × wave FE	0.069	8,154
	(0.045)	

Notes: Every row is the preferred specification – cohort × sector fixed effects, survey-wave and region fixed effects and the baseline control vector – with the stated restriction added. Row 2 adds a full set of industry × survey-wave interactions, which absorb any shock common to an industry in a given year and therefore do not require identifying in advance which industries boomed. Row 3 restricts both sectors to services, where most public employment is, so that public employees are compared with private employees in similar activities. Row 4 removes industry and construction. Rows 5 and 6 exclude Western region, where offshore oil production began in December 2010, and then also Ashanti, the centre of gold mining. Row 7 replaces region and wave effects with region × wave effects, absorbing any local shock. The industry variable is coarse and is missing for part of the sample, so rows that condition on it rest on fewer observations. Standard errors clustered by cohort.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.